Strategic Under-Specification: A Stackelberg Game Between User and Assistant

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Abstract

Users of coding assistants routinely write terse prompts, relying on the assistant to resolve missing details through guessing or clarifying questions. We develop a contract-theoretic model of endogenous clarification in human–AI interaction. Prompt shortening reflects rational effort substitution ($m^*(a) \leq m^*(0)$) rather than inefficient moral hazard: by the envelope theorem, autonomous model improvements weakly improve user welfare despite shortening prompts. When clarification policies are opaque, naive users who overestimate the assistant's willingness to ask under-specify relative to what true operational policies require, incurring quadratic welfare losses. Conversely, under transparent Stackelberg commitment, an assistant with asking friction bias ($\lambda_A > c_Q$) suppresses clarification, inducing sophisticated users to defensively over-specify upstream ($m^* > m^{FB}$) to insure against unaddressed defects.

Closing the loop under uniform *pooling* and effort-conditioned *screening* governance regimes, we characterize equilibrium clarification and welfare. Under pooling, provider asking friction suppresses clarification, generating an equilibrium capability reversal (the "guessing trap"): when the assistant's asking rate declines sufficiently steeply with autonomous accuracy, prompt lengths increase and user welfare falls. Under screening menus, provider bias induces adverse-selection downward distortion on high-cost terse users ($a_H^{SB} < a_H^B$), while boundary relaxation moderates this distortion. Furthermore, we show that under mild bias, pooling can deliver strictly higher user welfare than screening because a biased screening assistant exploits contractible specification to dictate inefficiently high prompt effort. We delineate genuine adverse-selection distortion from technological clarification shutdown, characterize conditions under which the unclipped closed forms hold versus boundary clipping, and formulate a pre-registered experimental design with statistical power analysis for human-subject testing.

Keywords: Human–AI Interaction, Clarification Policies, Stackelberg Games, Contract Theory, Adverse Selection, Delegation.

Code Availability: Simulation suite and verification scripts are available at `https://github.com/pranjal-0802/strategic-underspecification`.

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1 Introduction

A user typing a coding request faces a fundamental economic trade-off: spend cognitive effort articulating edge cases, naming conventions, and constraints, or leave them implicit and trust the assistant to infer them. The assistant, in turn, must decide for each unspecified detail whether to guess or to ask. Neither choice is made in a vacuum: the user's specification effort depends on what they expect the assistant to do with silence, and the assistant's ask/guess policy is chosen

(explicitly, or as an emergent property of training and product incentives) with some implicit model of how users will respond.

This paper provides a mechanism-design theory of endogenous clarification in AI assistance. Its central premise is that prompt shortening reflects rational effort substitution in response to automated clarification ($m^*(a) \leq m^*(0)$), while welfare-reducing under-specification arises when users naively anticipate more clarification than the assistant provides ($m^*_{\text{naive}} < m^*_{\text{soph}}$). When the assistant's policy is transparently chosen as a Stackelberg leader, provider asking friction ($\lambda_A > c_Q$) suppresses clarification, forcing users to over-specify relative to first-best ($m^* > m^{FB}$) and generating an equilibrium capability trap where better guessing can perversely lengthen prompts and lower welfare. Section 2 begins with a decision-theoretic benchmark where the assistant's policy is exogenous, proving that improved guessing accuracy alone weakly improves user welfare despite shortening prompts. Section 3 formalizes effort substitution and belief-driven under-specification under opaque policies. Sections 4–7 close the loop by modeling the assistant as a Stackelberg leader anticipating user best responses under pooling and screening instruments. Section 8 translates the equilibrium characterizations into falsifiable behavioral predictions. Section 10 evaluates exact nonlinear robustness and discusses institutional extensions.

1.1 Related literature

This work bridges contract theory and the emerging literature on formal models of human–AI interaction.

Contract theory and screening without transfers. Our screening framework builds on classical adverse-selection and mechanism design models (Mirrlees, 1971; Myerson, 1979, 1981; Baron and Myerson, 1982; Mussa and Rosen, 1978; Laffont and Tirole, 1986) that exploit the Spence–Mirrlees single-crossing property (Spence, 1973) to separate heterogeneous agents. However, unlike classical monopoly or regulatory screening where the principal uses monetary transfers to extract information rents, here the assistant screens strictly through an interactive service margin (the ask rate a offered in response to specification effort m). This connects to non-monetary screening and costly verification (Townsend, 1979; Ben-Porath et al., 2014), where information revelation is governed by operational friction rather than cash transfers.

Delegation, authority, and non-monetary mechanism design. Our analysis relates closely to delegation and non-monetary mechanism design (Holmström, 1984; Dessein, 2002; Alonso and Matouschek, 2008; Armstrong and Vickers, 2010; Ambrus and Egorov, 2017), where a principal designs an optimal decision menu without monetary transfers, and to models of formal versus real authority (Aghion and Tirole, 1997), where an agent's upstream effort is shaped by who retains control over downstream execution. In our setting, leaving attributes unspecified implicitly delegates their resolution to the assistant. Furthermore, because the assistant's asking policy determines which state dimensions are actively resolved, our framework shares conceptual foundations with Bayesian persuasion and strategic information design (Kamenica and Gentzkow, 2011), while focusing on the effort-substitution trade-off in prompt articulation.

Costly communication and strategic disclosure. The user's specification decision formalizes costly disclosure under strategic interpretation, connecting to strategic information transmission (Crawford and Sobel, 1982), voluntary quality disclosure (Grossman, 1981), and rational inattention (Sims, 2003). In our setting, under-specification arises not from an incentive to deceive, but from the user's rational economizing on specification costs when anticipating automated assistance.

Assistance games and human–AI interaction. Our bi-level formulation models the interaction as a Stackelberg game (von Stackelberg, 1934), connecting conceptually to the literature on assistance games and cooperative inverse reinforcement learning (Hadfield-Menell et al., 2016), as well as incomplete contracting in AI alignment (Hadfield-Menell and Hadfield, 2019), where an autonomous assistant acts to maximize human utility under incomplete specification. However, while assistance games formalize the assistant's uncertainty over the human's underlying *objective or reward function*, in our setting the economic task objective is known and shared (downstream success value V and error loss L). Instead, the informational asymmetry resides entirely in the user's private *specification cost* κ. The strategic margin in our model is therefore not reward learning or preference alignment, but rather the contract-theoretic trade-off between prompt specification effort and automated clarification under costly communication. Our model provides a contract-theoretic microfoundation for this ask-versus-guess margin under asymmetric specification costs.

Clarification, ambiguity, and RLHF incentives. A parallel literature in natural language processing investigates algorithms for selective clarification under ambiguity (Rao and Daumé III, 2018; Aliannejadi et al., 2019; Zamani et al., 2020; Min et al., 2020; Kuhn et al., 2023). Recent evaluations show that state-of-the-art language models often guess prematurely under ambiguity rather than asking clarifying questions (Laban et al., 2025). In conversational AI, this premature guessing is reinforced by reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO), which systematically favor immediate answers and introduce conversational length and sycophancy biases (Sharma et al., 2023; Singhal et al., 2023). Existing engineering studies treat prompts as exogenous static benchmarks. Our framework endogenizes prompt specification effort as an equilibrium best response to provider clarification policies and asking frictions.

2 Environment and the Decision-Theoretic Benchmark

2.1 Setup

A task has k binary, independent, intent-relevant attributes $\theta = (\theta_1, \ldots, \theta_k) \in \{0,1\}^k$, each drawn i.i.d. uniform and privately known to the user. The user chooses a specification effort level $m \in \{0, \ldots, k\}$ and truthfully reveals m attributes.[1] Specifying m attributes costs $c(m)$, with $c(0) = 0$, $c' \geq 0$, $c'' \geq 0$.

For each of the $k - m$ unspecified attributes, the assistant either *asks* (revealing the true value, at a cost to the user) or *guesses*, succeeding independently with probability $g \in (0,1)$. We model the assistant's policy at this stage as a uniform ask rate $a \in [0,1]$ applied to every unspecified attribute, so each is correctly resolved with probability $q(a,g) = a + (1-a)g$. Task success requires *all* k attributes to be correctly resolved (conjunctive: one wrong parameter or convention typically breaks the deliverable). The user's payoff is

$$U(m; a, g) \;=\; V\, q(a,g)^{\,k-m} \;-\; c(m) \;-\; a\, c_Q\, (k-m),$$

with $V > 0$ the value of success and $c_Q > 0$ the user's cognitive cost of answering one clarifying question. We treat m as continuous on $[0,k]$ for tractability.

[1] We assume truthful revelation conditional on the choice to specify: the strategic margin is *how much* to say, not whether to lie about what is said. A cheap-talk layer on top of this costly-specification layer is a natural extension; see Section 10.

2.2 Proposition 1: capability improvements and effort substitution

Proposition 1. *Fix a. Let $U^*(g) = \max_{m\in[0,k]} U(m; a, g)$. Then U^* is weakly increasing in g.*

Proof. For any fixed $m \in [0, k]$, the user's objective $U(m; a, g) = Vq(a, g)^{k-m} - c(m) - ac_Q(k - m)$ is pointwise non-decreasing in g because $\partial U/\partial g = V(k - m)q(a, g)^{k-m-1}(1 - a) \geq 0$. As the upper envelope of a family of weakly increasing functions, the maximized utility $U^*(g) = \max_{m\in[0,k]} U(m; a, g)$ is weakly increasing in g directly, without requiring differentiability. Under differentiability, the envelope theorem confirms that at the interior optimum $m^*(g)$,

$$\frac{dU^*}{dg} \;=\; \left.\frac{\partial U}{\partial g}\right|_{m=m^*(g)} \;=\; V(k - m^*)q^{k-m^*-1}(1 - a) \;\geq\; 0.$$

The envelope theorem lets us ignore dm^*/dg: because m^* is chosen optimally, its first-order response to g has no first-order effect on U^*. □

Assumption 1 (Regularity)**.** *In the relevant range of m, $(k - m) \ln q(a, g) \geq -1$.*

Remark 1 (Interpreting Assumption 1 and Model Concavity)**.** *The regularity condition $(k - m) \ln q(a, g) \geq -1$ provides a sufficient condition for decreasing differences in (m, g), which guarantees monotone comparative statics via Topkis's theorem (Topkis, 1978, 1998; Milgrom and Shannon, 1994). The direction is pinned down by the cross-partial of utility: for fixed a,*

$$\frac{\partial^2 U}{\partial m\, \partial g} \;=\; -V(1 - a)\, q^{\,k-m-1}\Big[(k - m) \ln q + 1\Big].$$

Thus U exhibits decreasing differences in (m, g) whenever $(k - m) \ln q + 1 \geq 0$, which is precisely Assumption 1. Because this condition is most demanding at $m = 0$, global Topkis submodularity on the entire domain $[0, k]$ requires $k \ln q \geq -1$. When k is large or baseline accuracy is modest (e.g., $k = 10, g = 0.5$), this global condition is violated. However, in optimizing behavior, users articulate the bulk of attributes upstream, keeping residual attributes $k - m^*$ small and ensuring that the local decreasing differences condition $(k - m^*) \ln q \geq -1$ holds across nearly the entire parameter space (verified in 89.9% of grid configurations, with empirical monotonicity surviving in 100% of tested domains; see Section 10). If this condition fails locally, U exhibits increasing differences, and higher guessing accuracy can locally induce higher specification effort before falling as $q \to 1$.

We also note that in the exact conjunctive model, the gross benefit Vq^{k-m} is strictly convex in m (since $\partial^2[Vq^{k-m}]/\partial m^2 = Vq^{k-m}(\ln q)^2 > 0$). Global concavity of U in m therefore requires the specification cost curvature $c''(m) = \kappa$ to strictly exceed $Vq^{k-m}(\ln q)^2$ across $[0, k]$. Under the linear-risk specification introduced in Section 4, by contrast, gross benefit is linear in m, guaranteeing that U is globally strictly concave in m for all $\kappa > 0$.

Corollary 1. *Under Assumption 1, $m^*(g)$ is weakly decreasing in g (decreasing differences in (m, g); Topkis's theorem).*

Proof. Immediate from the cross-partial computed in Remark 1: Assumption 1 is exactly the condition $\partial^2 U/\partial m\, \partial g \leq 0$ on the relevant range, which is decreasing differences in (m, g); Topkis's theorem then gives $m^*(g)$ weakly decreasing. □

Thus, holding the assistant's policy fixed, users specify less as the assistant becomes more adept at guessing, yet they are never worse off for doing so. Proposition 1 serves primarily as a baseline: it applies the envelope theorem to a single agent optimizing against a technology parameter that enters

the payoff with a non-negative derivative, a property that holds generally in models of this class. Its purpose is not to present a surprising result, but to establish the rigorous benchmark against which claimed welfare loss from prompt shortening must be evaluated, ensuring that Proposition 2 isolates the true source of strategic inefficiency. This frames the central question of this paper: what happens once the policy a is no longer exogenous, but chosen by a strategic assistant that anticipates the user's response?

3 Sources of Strategic Under-Specification

Proposition 1 assumed the assistant's ask rate is fixed and that users correctly anticipate $q(a, g)$. Both assumptions can fail.

3.1 Cost externality

If a third party (such as a code reviewer or downstream production system) bears part of the cost of an erroneous guess, the user's private parameters (κ, V) understate the social costs, causing the user to under-specify relative to the social optimum at any fixed g. Under linear risk with external defect loss $L_{\text{soc}} > L$, the gap between social and private specification effort is $(L_{\text{soc}} - L)(1 - g)/\kappa$, which contracts as guessing accuracy g improves because autonomous errors become less frequent. While this constitutes a standard Pigouvian externality, our primary focus lies in strategic and informational friction.

3.2 Miscalibrated belief updating under silent failure

If incorrect guesses are not always detected, and the detection probability $d(m)$ is increasing in m, users infer g from experience rather than observing it directly. Under-specifying users are consequently less likely to catch their own errors, generating an upward-biased estimate $\hat{g}$ that reinforces under-specification. This mechanism represents a dynamic learning process rather than a static equilibrium outcome; we formalize its testable implications in Section 8 (prediction P5).

3.3 Belief about the assistant's policy and equilibrium under-specification

When the clarification policy a is endogenous, the user's specification decision depends on their belief regarding the assistant's asking rate. Let users be *sophisticated* (correctly inferring the assistant's true operational ask rate a_A) or *naive* (believing the assistant follows a higher clarification rate $a^\dagger > a_A$, such as the unconstrained welfare-optimal policy).

Definition 1 (Effort Substitution and Belief-Driven Under-Specification)**.** *A user interacting with an automated assistant modifies upstream specification effort m across two distinct economic margins:*

(i) ***Rational effort substitution:*** *$m^*(a) \leq m^*(0)$, where the user economizes on prompt articulation by delegating attribute resolution to automated assistance; and*

(ii) ***Belief-driven under-specification:*** *$m^*(\hat{a}) < m^*(a)$, where a user holding optimistic or miscalibrated beliefs $\hat{a} > a$ under-specifies relative to the assistant's true operational ask rate.*

In human–AI interaction, varying the clarification rate a engages two opposing economic channels:

1. *The accuracy channel:* A higher ask rate increases effective resolution accuracy $q(a, g) = a + (1 - a)g$. Under decreasing differences (Assumption 1), greater accuracy reduces the marginal value of upstream prompt effort, pushing specification downward.

2. *The question-cost channel:* Each clarifying question imposes an interactive answering cost c_Q on the user. Because the total answering cost $ac_Q(k - m)$ is decreasing in specification m, a higher ask rate increases the marginal penalty of leaving attributes unspecified, pushing specification upward (pre-emption).

Under the linear-risk specification analyzed in Sections 4–6, where the net benefit of clarification $s \equiv \Lambda - c_Q > 0$ is positive, the cross-partial derivative of user utility is:

$$\frac{\partial^2 U}{\partial m \, \partial a} \;=\; -(\Lambda - c_Q) \;<\; 0.$$

The accuracy channel strictly dominates the question-cost channel, ensuring that U exhibits strictly decreasing differences in (m, a).

Proposition 2 (Prompt Under-Specification under Naive Beliefs)**.** *Under the linear-risk specification with $\Lambda > c_Q$, naive users choose weakly lower specification effort than sophisticated users:*

$$m^*_{naive} \;\leq\; m^*_{soph},$$

holding strictly ($m^*_{naive} < m^*_{soph}$) for all interior types satisfying $\kappa > \frac{\Lambda(1-a^\dagger)+a^\dagger c_Q}{k}$. Consequently, naive users achieve weakly lower true welfare, $U(m^*_{naive}; a_A) \leq U(m^*_{soph}; a_A)$, strictly so on the interior domain. Holding raw guessing accuracy g fixed, naive users under-specify relative to the assistant's true operational policy and suffer an equilibrium welfare loss.

Proof. Under linear risk on the physical domain $m \in [0, k]$, the follower's best response to an anticipated ask rate a is $m^*(\kappa; a) = \min\{k, \beta(\kappa; a)\}$, where $\beta(\kappa; a) \equiv \frac{\Lambda(1-a)+ac_Q}{\kappa}$. Since $\Lambda > c_Q$, the unconstrained response $\beta(\kappa; a)$ is strictly decreasing in a: $\partial\beta/\partial a = (c_Q - \Lambda)/\kappa < 0$. A naive user anticipating $a^\dagger > a_A$ therefore satisfies $\beta(\kappa; a^\dagger) < \beta(\kappa; a_A)$. For low-cost types with $\kappa \leq \frac{\Lambda(1-a^\dagger)+a^\dagger c_Q}{k}$, we have $\beta(\kappa; a^\dagger) \geq k$, so $\beta(\kappa; a_A) > k$ as well. Both beliefs therefore yield boundary saturation: $m^*_{\text{naive}} = m^*_{\text{soph}} = k$, and true welfare is identical ($U(m^*_{\text{naive}}; a_A) = U(m^*_{\text{soph}}; a_A)$). For all interior types satisfying $\kappa > \frac{\Lambda(1-a^\dagger)+a^\dagger c_Q}{k}$, we have $\beta(\kappa; a^\dagger) < k$, which guarantees:

$$m^*_{\text{naive}} \;=\; \beta(\kappa; a^\dagger) \;<\; \min\{k, \beta(\kappa; a_A)\} \;=\; m^*_{\text{soph}}.$$

Because true utility $U(m; a_A)$ is strictly concave in m with unique global maximizer on $[0, k]$ at m^*_{soph}, any choice $m \neq m^*_{\text{soph}}$ incurs a strict utility loss. On the fully interior domain where $m^*_{\text{soph}} = \beta(\kappa; a_A) < k$, evaluating the quadratic shortfall directly yields:

$$U(m^*_{\text{soph}}; a_A) - U(m^*_{\text{naive}}; a_A) \;=\; \frac{\kappa}{2}\big(m^*_{\text{soph}} - m^*_{\text{naive}}\big)^2 \;=\; \frac{(\Lambda - c_Q)^2}{2\kappa}\,(a^\dagger - a_A)^2 \;>\; 0,$$

confirming that interior naive users are strictly worse off.

Notice that on the interior domain, the under-specification gap is:

$$m^*_{\text{soph}} - m^*_{\text{naive}} \;=\; \frac{(\Lambda(g) - c_Q)(a^\dagger - a_A)}{\kappa} \;=\; \frac{(L(1-g) - c_Q)(a^\dagger - a_A)}{\kappa}.$$

Differentiating with respect to raw guessing accuracy g yields:

$$\frac{\partial(m^*_{\text{soph}} - m^*_{\text{naive}})}{\partial g} \;=\; -\,\frac{L(a^\dagger - a_A)}{\kappa} \;<\; 0.$$

As the assistant's autonomous guessing capability improves, the distortion gap between naive and sophisticated users strictly contracts, because guessing errors become less frequent and the economic penalty of misplaced trust declines. □

Remark 2 (Linear Risk Foundation and Non-Linear Cross-Partial Reversal)**.** *The monotone under-specification result of Proposition 2 relies directly on the linear-risk structure, where the cross-partial derivative is identically $\partial^2 U/\partial m\,\partial a = -(\Lambda - c_Q) < 0$. Under linear risk, whenever clarification is cost-effective ($\Lambda > c_Q$), user utility exhibits decreasing differences globally, ensuring that anticipated asking strictly reduces prompt effort ($m^*(\hat{a}) < m^*(a)$ for $\hat{a} > a$). In contrast, in the exact conjunctive model $U(m, a) = Vq(a, g)^{k-m} - c(m) - ac_Q(k - m)$, the cross-partial derivative is:*

$$\frac{\partial^2 U}{\partial m\,\partial a} \;=\; -V(1-g)\,q^{k-m-1}\big[(k-m)\ln q + 1\big] \;+\; c_Q.$$

Here, varying the ask rate activates two opposing forces: the accuracy benefit $-V(1-g)q^{k-m-1}[(k-m)\ln q + 1] \le 0$ and the interactive question-answering friction $+c_Q > 0$. When question costs are large relative to task stakes or when autonomous accuracy is high (e.g., $k = 5, V = 10, \kappa = 0.5, g = 0.8, c_Q = 2$, where $\Lambda = V(1-g) = 2.0 = c_Q$, so net asking surplus $s = 0$), the question-cost channel dominates. In that case, asking provides no net benefit ($U(0) > U(0.1)$), and users anticipating higher asking choose higher upstream effort ($m^*_{naive} = 3.92 > 3.15 = m^*_{soph}$) to pre-empt conversational interruptions. Proposition 2 thus isolates the economic environment where clarification provides positive net value ($\Lambda > c_Q$), which represents the canonical operating regime for helpful automated assistants.

This identifies the dual nature of specification distortions across informational regimes:

1. *Opaque policies and strategic under-specification:* When users naively trust the assistant to provide first-best clarification ($a^\dagger = a^{FB} = 1$), they write concise prompts calibrated to full assistance ($m^*_{\text{naive}} = m^{FB} = c_Q/\kappa$). If the assistant is a biased Stackelberg leader implementing an asking-suppressed policy $a_A = a^{SE} < 1$, naive users under-specify relative to what the biased environment demands ($m^*_{\text{naive}} < m^*_{\text{soph}}$). Their welfare shortfall from misplaced trust is:

$$\Delta U_{\text{naive}} \;\equiv\; U(m^*_{\text{soph}}; a^{SE}) \;-\; U(m^*_{\text{naive}}; a^{SE}) \;=\; \frac{(\Lambda - c_Q)^2}{2\kappa}\big(1 - a^{SE}\big)^2 \;=\; \frac{s^2(1-a^{SE})^2}{2\kappa},$$

   which is strictly positive whenever the leader strategically withholds clarification ($a^{SE} < 1$).

2. *Transparent policies and defensive over-specification:* Under transparent Stackelberg commitment, sophisticated users know the leader's operational policy a^{SE}. Because the biased leader under-asks ($a^{SE} < 1$), sophisticated users cannot rely on clarification downstream. They compensate by defensively lengthening prompts beyond the first-best level ($m^*(a^{SE}) > m^{FB}$), substituting costly private effort for missing automated assistance.

Sections 4–7 close the loop by modeling how the assistant endogenously chooses this clarification policy under pooling and screening governance mechanisms.

4 The Assistant's Policy as a Stackelberg Game

4.1 General formulation

The assistant (leader) commits ex ante to a policy π before meeting any user. A population of users (followers) has privately known specification-cost type $\kappa \sim F$ on $[\kappa_L, \kappa_H]$, with cost $c(m;\kappa) = \kappa m^2/2$, and best-responds $m^*(\kappa;\pi) = \arg\max_m U(m;\pi,\kappa)$. The leader's own objective Π need not equal U: we write

$$\Pi_\kappa(m,a) \;=\; \mu_A\, U(m,a;\kappa) \;-\; (\lambda_A - c_Q)\, a\,(k-m),$$

where $\mu_A \in (0,1]$ discounts the leader's weight on true downstream user welfare, c_Q is the user's true cognitive cost of answering one clarifying question, and λ_A is the leader's own perceived or institutional cost per clarifying question. In commercial AI systems, misalignment along these margins arises from concrete engineering and economic forces:

1. *Asking friction bias* ($\lambda_A > c_Q$): Reinforcement learning from human feedback (RLHF) and preference optimization algorithms (such as DPO) commonly evaluate conversation turns in isolation. Reward models systematically penalize "evasive" clarifying questions in favor of immediate code generation because rating pools favor direct answers, reinforcing sycophancy and conversational turn penalties (Sharma et al., 2023; Singhal et al., 2023; Laban et al., 2025). Furthermore, multi-turn interactions increase token costs, context-window expansion, and inference latency, creating provider-side economic pressure to suppress clarification turns.

2. *Downstream welfare discounting* ($\mu_A < 1$): Turn-level user satisfaction signals (such as prompt acceptance or immediate positive feedback) capture immediate session fluency. In contrast, the true economic loss L from broken conventions, subtle edge-case errors, or security defects is typically discovered hours or days later in downstream execution, test suites, or code review. Consequently, training objectives tuned to immediate session retention tend to discount downstream correctness.

When asking friction is aligned ($\lambda_A = c_Q$), the leader's objective satisfies $\Pi_\kappa \equiv \mu_A U(\cdot;\kappa)$, sharing the identical preferences and policy maximizers as true user welfare for any positive welfare weight $\mu_A > 0$. Because user utility U already incorporates private questioning cost $-ac_Q(k-m)$, expanding the leader's objective yields:

$$\Pi_\kappa(m,a) \;=\; \mu_A\left[V \;-\; \Lambda(1-a)(k-m) \;-\; \frac{\kappa}{2}m^2\right] \;-\; \Big(\lambda_A - (1-\mu_A)c_Q\Big)\, a\,(k-m).$$

The total effective weight multiplying clarification volume $-a(k-m)$ in Π_κ is therefore $\lambda_A - (1-\mu_A)c_Q$, which simplifies directly to λ_A in the benchmark case $\mu_A = 1$. Downstream welfare discounting $\mu_A < 1$ creates policy distortion through interaction with excess asking friction ($\lambda_A \neq c_Q$), inflating the effective friction per unit of altruism $(\lambda_A - c_Q)/\mu_A$. The leader solves $\pi^* = \arg\max_\pi \mathbb{E}_{\kappa\sim F}[\Pi_\kappa(m^*(\kappa;\pi),\pi)]$, a bilevel program.

Remark 3 (Existence)**.** *For Model I below ($\pi = a \in [0,1]$, a compact scalar policy), $m^*(\kappa;a)$ is continuous in a by Berge's Maximum Theorem (since U is strictly concave in m on the compact domain $[0,k]$), so the expected leader objective $\mathbb{E}_\kappa[\Pi_\kappa(m^*(\kappa;a),a)]$ is continuous on $[0,1]$ and a maximizer exists by Weierstrass's Extreme Value Theorem. For Model II below (a finite menu over two types), the feasible allocation space is a non-empty, compact subset of $[0,k]^2 \times [0,1]^2$ defined by continuous (bilinear in (m,a) and quadratic in m) IC and IR constraints. The constraint set is non-empty because the degenerate pooling contract $(m^*(a),a,m^*(a),a)$ is always incentive compatible. By Weierstrass's theorem, the continuous leader objective attains a global maximum on this compact set.*

For tractability in what follows we evaluate the linear-risk objective:

$$U(m, a; \kappa) \;=\; V \;-\; \Lambda(1-a)(k-m) \;-\; \frac{\kappa}{2}m^2 \;-\; a\,c_Q\,(k-m),$$

where $\Lambda \equiv L(1-g)$. This specification admits two complementary economic microfoundations:

1. *Additive independent defects ($kL \leq V$):* Each unaddressed attribute independently produces a localized defect cost $L \leq V/k$. Total project value is destroyed only if defects accumulate across attributes. Our benchmark calibration sets $k = 10, L = 10, V = 100$, where full failure ($kL = 100$) exhausts task value V.

2. *Systemic bottleneck conjunctive model ($L = V$):* The project succeeds only if all attributes are defect-free: $Vq(a,g)^{k-m}$. A first-order Taylor expansion around $q \to 1$ yields $Vq^{k-m} \approx V - V(1-q)(k-m)$, which corresponds to setting per-attribute stake $L = V = 100$.

Throughout Sections 4.2–6, we maintain the physical feasibility constraint $m \in [0, k]$, so that followers cannot specify more attributes than exist in the task. When unconstrained first-order conditions yield $m^*(\kappa; a) = \frac{\Lambda(1-a)+ac_Q}{\kappa} > k$ (which occurs for low-cost types with $\kappa < \Lambda/k$), best-response effort saturates at the physical boundary $m^* = k$.

4.2 The first-best benchmark and physical feasibility

Before analyzing either strategic regime, we characterize the fully-informed, fully-aligned social optimum:

$$(m^{FB}(\kappa), a^{FB}(\kappa)) \;=\; \arg \max_{m \in [0,k],\, a \in [0,1]} U(m, a; \kappa).$$

Proposition 3 (First-Best Policy on the Feasible Domain)**.** *On the physically feasible domain $m \in [0, k]$, under the linear-risk specification with $\Lambda > c_Q$:*

(i) Asking weakly dominates guessing for every cost type $\kappa > 0$: $a^{FB}(\kappa) = 1$, with optimal specification $m^{FB}(\kappa) = \min\{k, c_Q/\kappa\}$.

(ii) For any type with $\kappa > c_Q/k$, asking strictly dominates guessing: $U(m^*(1), 1) > U(m^*(0), 0)$. For boundary types with $\kappa \leq c_Q/k$, specification effort saturates at $m^{FB} = k$, leaving zero residual attributes ($k - m = 0$) and rendering the asking policy payoff-irrelevant.

Proof. For any fixed specification effort $m \in [0, k]$, the partial derivative of utility with respect to a is:

$$\frac{\partial U}{\partial a} \;=\; (\Lambda - c_Q)(k-m).$$

Because $\Lambda > c_Q$, this derivative is strictly positive for all $m < k$ and zero at $m = k$. Now consider the joint optimization over $m \in [0, k]$ and $a \in \{0, 1\}$. Under $a = 1$, the user chooses $m^*(1) = \min\{k, c_Q/\kappa\}$. Under $a = 0$, the user chooses $m^*(0) = \min\{k, \Lambda/\kappa\}$. We evaluate the three type intervals:

- If $\kappa \geq \Lambda/k$, both best responses remain interior: $m^*(1) = c_Q/\kappa \leq m^*(0) = \Lambda/\kappa \leq k$. Evaluating the welfare difference:

$$U(m^*(1), 1; \kappa) - U(m^*(0), 0; \kappa) \;=\; (\Lambda - c_Q)\left(k - \frac{\Lambda + c_Q}{2\kappa}\right).$$

  Since $\kappa \geq \Lambda/k > \frac{\Lambda+c_Q}{2k}$, the bracketed term satisfies $k - \frac{\Lambda+c_Q}{2\kappa} > k - k = 0$. Hence $U(m^*(1), 1; \kappa) > U(m^*(0), 0; \kappa)$ strictly.

- If $c_Q/k < \kappa < \Lambda/k$, unconstrained effort under $a = 0$ exceeds k, so specification under guessing saturates at the boundary $m^*(0) = k$, yielding $U(k, 0; \kappa) = V - \frac{\kappa}{2}k^2$. Under asking, effort remains interior at $m^*(1) = c_Q/\kappa < k$, yielding $U(m^*(1), 1; \kappa) = V - c_Q k + \frac{c_Q^2}{2\kappa}$. Subtracting:

$$U(m^*(1), 1; \kappa) - U(k, 0; \kappa) \;=\; \frac{\kappa k^2}{2} - c_Q k + \frac{c_Q^2}{2\kappa} \;=\; \frac{(\kappa k - c_Q)^2}{2\kappa} \;>\; 0.$$

- If $\kappa \leq c_Q/k$, specification effort saturates at $m^* = k$ under both policies, so $U(k, 1; \kappa) = U(k, 0; \kappa) = V - \frac{\kappa}{2}k^2$.

Therefore, $a^{FB}(\kappa) = 1$ is globally optimal across all types $\kappa > 0$, with indifference only on the boundary set $\kappa \leq c_Q/k$ where $m^{FB} = k$. □

Remark 4 (The Unclipped Threshold κ^* and Genuine Asking–Guessing Trade-offs)**.** *In unconstrained algebra without the feasibility constraint $m \leq k$, setting the difference $(\Lambda - c_Q)(k - \frac{\Lambda + c_Q}{2\kappa})$ to zero yields the formal threshold $\kappa^* \equiv \frac{\Lambda + c_Q}{2k}$. However, since $\Lambda > c_Q$, we have $\kappa^* < \Lambda/k$ strictly. The region $\kappa < \kappa^*$ where guessing formally appeared to beat asking in unclipped algebra lies entirely in the unphysical domain where unconstrained effort Λ/κ exceeds k (the user would specify more attributes than exist in the task). Once $m \leq k$ is enforced, asking weakly dominates guessing everywhere under linear risk because asking resolves uncertainty at lower unit cost ($c_Q < \Lambda$) while allowing the user to economize on specification effort.*

Genuine interior thresholds where $a^{FB} \in \{0, 1\}$ strictly partitions the type space arise from two natural richer specifications:

1. Exact conjunctive success: *Under $Vq(a, g)^{k-m} - c(m) - ac_Q(k - m)$, the marginal gain from clarification is $(k - m)[V(1 - g)q^{k-m-1} - c_Q]$. When baseline accuracy g is high or $k - m$ is small, this marginal gain turns negative, making guessing strictly optimal for low-cost types who specify most attributes upstream.*

2. Heterogeneous attribute stakes: *If attributes vary in importance with stakes Λ_i, efficiency prescribes asking only on high-stakes attributes ($\Lambda_i > c_Q$) and guessing on low-stakes attributes ($\Lambda_i < c_Q$).*

5 Model I: A Pooling (Scalar) Policy

The leader is restricted to a single population-wide ask rate $a \in [0, 1]$ and cannot condition on user-specific attributes or observed effort. This provides a natural benchmark for policies embedded uniformly at training time or applied without user-level profiling.

5.1 User best response

The FOC $\partial U/\partial m = \Lambda(1 - a) - \kappa m + ac_Q = 0$ gives, with $\beta(\kappa) = \Lambda/\kappa$ and $\gamma(\kappa) = (\Lambda - c_Q)/\kappa$,

$$m^*(\kappa; a) \;=\; \beta(\kappa) - a\,\gamma(\kappa), \qquad \frac{\partial m^*}{\partial a} \;=\; -\gamma(\kappa) \;=\; \frac{c_Q - \Lambda}{\kappa}.$$

We assume interior specification levels $m^* \in (0, k)$; for low-cost types where unconstrained $\beta(\kappa) \geq k$, specification effort saturates at $m^* = k$. The closed-form population moments $\bar{R}_0$ and $\bar{\gamma}$ below are exact when best responses remain interior, and serve as tractable linear approximations when boundary clipping binds; as verified in Section 10 and our computational robustness suite, the

qualitative regime structure and corner-solution properties are strictly preserved under boundary clipping and exact payoffs alike.

Remark 5 (Non-monotone effect of ask rate on specification)**.** *Unlike Corollary 1, the sign of $\partial m^*/\partial a$ here does not depend on type: it is positive for* every *type iff $c_Q > \Lambda$, i.e., iff answering a question costs users more than the expected loss it avoids. When that condition holds, a higher population ask rate induces users to specify* more*, not less; they pre-empt costly clarifying questions by articulating the answer directly. This introduces a new economic force relative to Section 2, where a operated solely through effective accuracy, and generates a distinct, directly testable prediction (prediction P1′ below).*

5.2 The leader's problem

Let $R(a,\kappa) = k - m^*(\kappa;a)$ denote the residual unspecified attributes. In the pooling regime, the assistant commits ex ante to an operational ask rate $a \in [0,1]$ applied population-wide, without observing or directly contracting on individual specification effort.

Evaluating the general leader objective from Section 4.1 at the follower's optimal best response $m^*(\kappa;a) = \beta(\kappa) - a\,\gamma(\kappa)$:

$$\Pi_\kappa(m^*(\kappa;a),a) \;=\; \mu_A\,U(m^*(\kappa;a),a;\kappa) \;-\; (\lambda_A - c_Q)\,a\,\big(k - m^*(\kappa;a)\big).$$

Substituting $U(m^*(\kappa;a),a;\kappa) = V - \Lambda(1-a)(k-m^*) - \frac{\kappa}{2}(m^*)^2 - a\,c_Q\,(k-m^*)$, using $k - m^*(\kappa;a) = R_0(\kappa) + a\,\gamma(\kappa)$, and defining the effective resolution stake $s \equiv \Lambda - c_Q$ and the assistant's excess asking friction $b \equiv \lambda_A - c_Q$, expanding in powers of a yields:

$$\Pi_\kappa(m^*(\kappa;a),a) \;=\; \mu_A\left(V - \Lambda k + \frac{\Lambda^2}{2\kappa}\right) \;+\; (\mu_A s - b)\,R_0(\kappa)\,a \;+\; \frac{(\mu_A s - 2b)\,\gamma(\kappa)}{2}\,a^2.$$

Averaging over $\kappa \sim F$ requires no moments beyond the population sufficient statistics already defined, $\bar R_0 = k - \mathbb{E}[\beta]$ and $\bar\gamma = \mathbb{E}[\gamma] = s\,\mathbb{E}[1/\kappa]$:

$$\Pi(a) \;=\; \text{const} \;+\; \alpha_1\,a \;+\; \alpha_2\,a^2,$$

where const $= \mu_A(V - \Lambda k) + \frac{\mu_A\Lambda^2}{2}\mathbb{E}[1/\kappa]$, and

$$\alpha_1 \;\equiv\; (\mu_A s - b)\,\bar R_0, \qquad \alpha_2 \;\equiv\; \frac{(\mu_A s - 2b)\,\bar\gamma}{2}.$$

Lemma 1 (Leader Objective as User Welfare with an Asking Friction Wedge)**.** *For any allocation (m,a), the leader's objective satisfies:*

$$\Pi(m,a) \;=\; \mu_A\Big[U(m,a) \;-\; \tilde\rho\,a\,(k-m)\Big],$$

where $\tilde\rho \equiv \frac{\lambda_A - c_Q}{\mu_A}$ is the normalized excess asking friction. The leader's objective incorporates true user welfare but introduces a per-question friction wedge $\tilde\rho$. In the leader's evaluation, clarifying questions are priced at an effective cost $c_Q^{eff} \equiv c_Q + \tilde\rho$. However, this is not an equivalence to a benevolent social planner, because followers continue to best-respond against their true private question cost c_Q, not c_Q^{eff}. Because $\mu_A > 0$ enters strictly as an overall positive scalar multiplier on the bracketed term, optimal leader policies depend on (μ_A, λ_A) solely through the excess friction ratio $\tilde\rho$.

Proposition 4 (Stackelberg-optimal pooling rate)**.** *Provided $\alpha_2 < 0$ (the second-order condition for strict concavity, equivalent to $2(\lambda_A - c_Q) > \mu_A(\Lambda - c_Q)$), the leader's unconstrained stationary point is*

$$a^{SE} \;=\; -\frac{\alpha_1}{2\alpha_2} \;=\; -\frac{(\mu_A s - b)\,\bar{R}_0}{(\mu_A s - 2b)\,\bar{\gamma}}.$$

When this stationary point lies in $[0,1]$, it is the unique global maximum of $\Pi(a)$ on $[0,1]$.

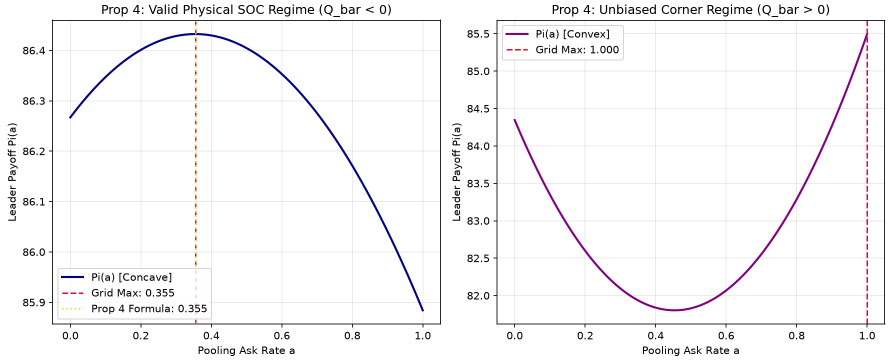


Figure 1: Leader pooling payoff $\Pi(a)$ and equilibrium ask rate a^{SE} under Model I, comparing the unclipped closed form against numerical grid optimization.

Corollary 2 (Friction bias suppresses the pooling ask rate on the interior branch)**.** *On the interior branch $\alpha_2 < 0$, differentiating the closed form in Proposition 4 directly with respect to λ_A (holding primitives $\mu_A, \Lambda, c_Q, \bar{R}_0, \bar{\gamma}$ fixed) gives:*

$$\frac{\partial a^{SE}}{\partial \lambda_A} \;=\; -\frac{\bar{R}_0}{\bar{\gamma}}\,\frac{\mu_A s}{(\mu_A s - 2b)^2} \;=\; -\frac{\bar{R}_0}{\bar{\gamma}}\,\frac{\mu_A(\Lambda - c_Q)}{\big[\mu_A(\Lambda - c_Q) - 2(\lambda_A - c_Q)\big]^2}.$$

For populations supported on the interior specification domain ($\kappa \geq \Lambda/k$ for all types in the support), follower effort satisfies $m^*(\kappa; 0) = \Lambda/\kappa \leq k$, ensuring that expected residual unspecified attributes under silence satisfy $\bar{R}_0 = k - \mathbb{E}[\beta] > 0$. Because $\bar{\gamma} > 0$, $\mu_A > 0$, and $s > 0$, it follows that:

$$\frac{\partial a^{SE}}{\partial \lambda_A} \;\leq\; 0.$$

On this interior domain, higher perceived clarification friction λ_A strictly suppresses the assistant's equilibrium asking rate. When the population includes lower-cost types ($\kappa < \Lambda/k$) whose effort saturates at $m^* = k$, the unclipped linear moment decomposition is modified by boundary clipping; numerical optimization across the clipped distribution confirms that higher friction λ_A continues to weakly suppress asking, transitioning monotonically from $a^{SE} = 1$ to $a^{SE} = 0$.

Corollary 3 (Continuous transition of the pooling ask rate across friction regimes)**.** *Under asking friction bias ($\lambda_A > c_Q$), the second-order coefficient satisfies $\alpha_2 < 0$ if and only if $\tilde{\rho} > s/2$. For a continuum population supported on the interior specification domain ($\kappa \geq \Lambda/k$, so that $\bar{R}_0 > 0$), the endpoint crossover threshold satisfies $\tilde{\rho}^* \equiv \frac{s(\bar{R}_0 + \bar{\gamma}/2)}{\bar{R}_0 + \bar{\gamma}} > s/2$ strictly. Consequently, on the entire convex branch ($\tilde{\rho} \leq s/2 < \tilde{\rho}^*$), the leader strictly prefers universal clarification ($\Pi(1) > \Pi(0)$), selecting $a^{SE} = 1$ unconditionally. On the strictly concave branch ($\tilde{\rho} > s/2$), the optimal pooling policy transitions continuously on $[0,1]$ across three distinct friction regimes:*

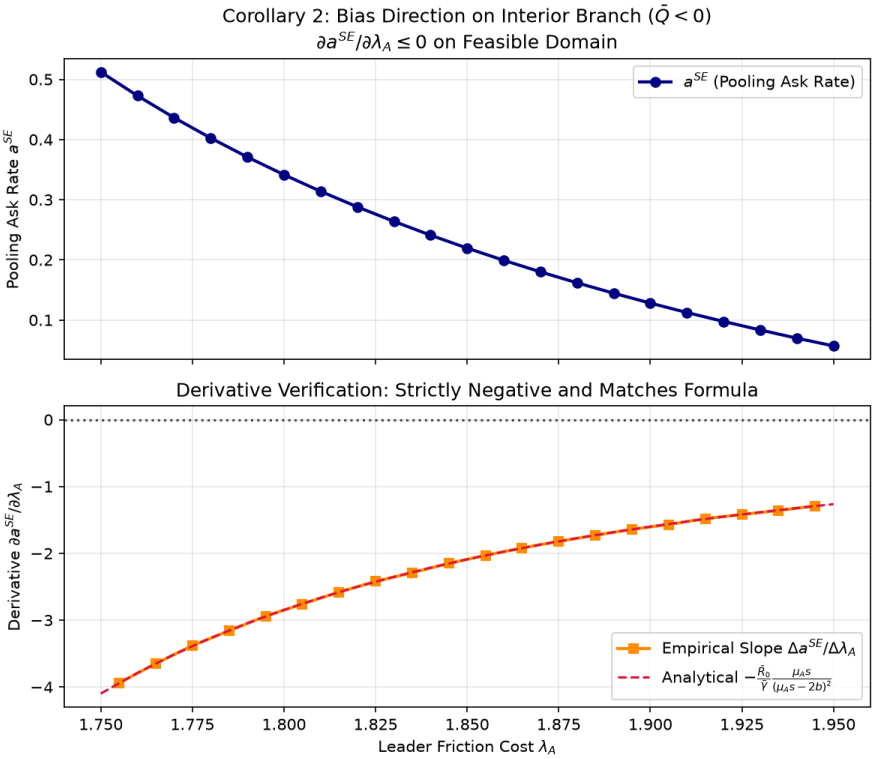


Figure 2: Perceived asking friction λ_A suppresses the Stackelberg pooling ask rate ($\partial a^{SE}/\partial \lambda_A \leq 0$) across varying altruism levels μ_A.

1. Mild bias ($\tilde{\rho} \leq \tilde{\rho}_1$): *For* $\tilde{\rho} \leq \tilde{\rho}_1 \equiv \frac{s(\bar{R}_0+\bar{\gamma})}{\bar{R}_0+2\bar{\gamma}}$, *the unconstrained stationary point satisfies* $-\alpha_1/(2\alpha_2) \geq 1$, *so the constrained maximum on* $[0,1]$ *is universal clarification:* $a^{SE} = 1$.

2. Moderate bias ($\tilde{\rho}_1 < \tilde{\rho} < s$): *The unconstrained stationary point is strictly interior, so the optimal ask rate is uniquely:*

$$a^{SE} \;=\; -\frac{\alpha_1}{2\alpha_2} \;=\; \frac{(s-\tilde{\rho})\bar{R}_0}{(2\tilde{\rho}-s)\bar{\gamma}} \;\in\; (0,1),$$

which is strictly decreasing in friction ($\partial a^{SE}/\partial\tilde{\rho} < 0$). *At the threshold value* $\tilde{\rho} = \tilde{\rho}^* \in (\tilde{\rho}_1, s)$, *endpoint payoffs satisfy* $\Pi(1) = \Pi(0)$, *which corresponds geometrically to the quadratic vertex lying exactly at* $a^{SE} = 1/2$. *The interior optimum transitions smoothly through this point without any regime discontinuity.*

3. Prohibitive friction ($\tilde{\rho} \geq s$, equivalently $c_Q^{eff} \geq \Lambda$): *The leader unconstrainedly prefers zero clarification* ($-\alpha_1/(2\alpha_2) \leq 0$), *shutting down asking completely:* $a^{SE} = 0$.

This continuous transition characterizes the single unclipped quadratic representation with continuous type support; in discrete type distributions (such as two-type populations $\kappa \in \{\kappa_L, \kappa_H\}$*) or under boundary clipping, piecewise best-response regimes can induce discrete policy shifts.*

Corollary 4 (Pooling is a corner solution in the unbiased case)**.** *In the unbiased case* ($\mu_A = 1, \lambda_A = c_Q \implies b = 0$), *the quadratic coefficient is:*

$$\alpha_2 \;=\; \frac{s\,\bar{\gamma}}{2} \;=\; \frac{s^2\,\mathbb{E}[1/\kappa]}{2} \;=\; \frac{(\Lambda - c_Q)^2\,\mathbb{E}[1/\kappa]}{2} \;\geq\; 0 \quad \textit{always.}$$

The second-order condition for an interior maximum therefore fails: $\Pi(a)$ *is weakly convex on* $[0,1]$, *its stationary point is a local minimum, and the true optimum is always a corner,* $a^{SE} \in \{0,1\}$, *not an interior compromise. Comparing the two corners directly in unclipped algebra:*

$$\Pi(1) - \Pi(0) \;=\; \alpha_1 + \alpha_2 \;=\; s\left(\bar{R}_0 + \frac{\bar{\gamma}}{2}\right) \;=\; (\Lambda - c_Q)\left[k - \frac{\Lambda + c_Q}{2}\,\mathbb{E}\left[\frac{1}{\kappa}\right]\right],$$

which yields the formal unclipped threshold $k^* \equiv \frac{\Lambda+c_Q}{2}\mathbb{E}[1/\kappa]$. *When follower best responses are physically constrained to the feasible domain* $m \leq k$, *Proposition 3 implies that asking* ($a = 1$) *weakly dominates guessing for every individual type in the support, so unbiased pooling under physical feasibility selects* $a^{SE} = 1$ *unconditionally. Genuine interior trade-offs between asking and guessing at the population level emerge under the exact conjunctive success model, as analyzed in Section 10.*

Proposition 5 (Equilibrium Reversal of Guessing Capability: The Guessing Trap)**.** *Under asking friction bias* ($\lambda_A > c_Q$), *an increase in assistant guessing capability* g *affects the interior pooling clarification rate according to:*

$$\frac{da^{SE}}{dg} \;<\; 0 \iff \mu_A \bar{R}_0(g) \;>\; (\mu_A s(g) - b)\,\mathbb{E}\left[\frac{1}{\kappa}\right]\left(1 - 2a^{SE}\right).$$

In particular:

(i) If $a^{SE} \geq 1/2$, *the condition holds unconditionally, so* $\frac{da^{SE}}{dg} < 0$. *If* $a^{SE} < 1/2$, *the derivative can reverse when* $\bar{R}_0$ *is small.*

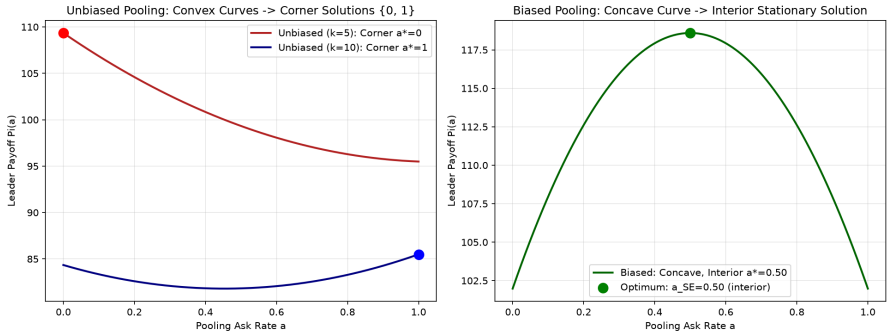


Figure 3: Payoff curvature across friction regimes: weakly convex corner solution in the unbiased regime ($\mu_A = 1, \lambda_A = c_Q$) versus strictly concave interior optimum under friction bias ($\lambda_A > c_Q$).

(ii) When $\frac{da^{SE}}{dg} < 0$*, prompt lengthening and welfare decline are mathematically equivalent and hold simultaneously if and only if*

$$-\frac{da^{SE}}{dg} > \frac{L(1-a^{SE})}{\Lambda(g) - c_Q},$$

reversing the comparative statics of Proposition 1 in Stackelberg equilibrium:

- ***Equilibrium prompt lengthening:*** $\frac{d\mathbb{E}[m^*]}{dg} > 0$.
- ***Equilibrium welfare decline:*** $\frac{dW}{dg} < 0$.

Proof. On the interior branch, the stationary point is $a^{SE}(g) = -\frac{\alpha_1(g)}{2\alpha_2(g)} = \frac{(\mu_A s(g) - b)\bar{R}_0(g)}{(2b - \mu_A s(g))\bar{\gamma}(g)}$, where $\Lambda(g) = L(1-g)$, $s(g) = L(1-g) - c_Q$, $\bar{R}_0(g) = k - \Lambda(g)\mathbb{E}[1/\kappa]$, and $\bar{\gamma}(g) = s(g)\mathbb{E}[1/\kappa]$. Differentiating with respect to g and collecting terms in $\mu_A, b, s(g), \bar{R}_0(g)$, and $\mathbb{E}[1/\kappa]$ yields:

$$\frac{da^{SE}}{dg} = -\frac{L}{\bar{\gamma}(g)(2b - \mu_A s(g))}\left[\mu_A \bar{R}_0(g) - (\mu_A s(g) - b)\mathbb{E}\left[\frac{1}{\kappa}\right]\left(1 - 2a^{SE}\right)\right].$$

Because the prefactor $-\frac{L}{\bar{\gamma}(g)(2b-\mu_A s(g))}$ is strictly negative on the concave interior branch ($2b > \mu_A s$), it follows that $da^{SE}/dg < 0$ if and only if $\mu_A \bar{R}_0(g) > (\mu_A s(g) - b)\mathbb{E}[1/\kappa](1 - 2a^{SE})$. When $a^{SE} \geq 1/2$, $(1 - 2a^{SE}) \leq 0$, ensuring the condition is satisfied unconditionally.

By the chain rule, total equilibrium follower effort satisfies:

$$\frac{dm^*}{dg} = \frac{\partial m^*}{\partial g} + \frac{\partial m^*}{\partial a}\frac{da^{SE}}{dg} = \frac{1}{\kappa}\left[-L(1-a^{SE}) - s(g)\frac{da^{SE}}{dg}\right].$$

Similarly, by the envelope theorem on follower choice of m^*, total equilibrium user welfare $W(g) \equiv \mathbb{E}[U(m^*(g), a^{SE}(g); g)]$ satisfies:

$$\frac{dW}{dg} = \mathbb{E}\left[\frac{\partial U}{\partial g}\bigg|_{m^*}\right] + \mathbb{E}\left[\frac{\partial U}{\partial a}\bigg|_{m^*}\right]\frac{da^{SE}}{dg} = \mathbb{E}[(k - m^*)]\left[L(1-a^{SE}) + s(g)\frac{da^{SE}}{dg}\right].$$

Because $\mathbb{E}[k - m^*] > 0$ and $\mathbb{E}[1/\kappa] > 0$, prompt lengthening ($d\mathbb{E}[m^*]/dg > 0$) and welfare decline ($dW/dg < 0$) are mathematically equivalent and hold simultaneously if and only if $-da^{SE}/dg > \frac{L(1-a^{SE})}{s(g)}$. □

Remark 6 (Analytical Conditions versus Clipped Numerical Calibration)**.** *Proposition 5 characterizes the unclipped interior branch where best responses remain interior ($m^*(\kappa) < k$). When $a^{SE} < 1/2$, the derivative condition can reverse; for example, in open-domain tasks with multiple alternatives where guessing accuracy satisfies $g \in [0.05, 0.10]$ (with $k = 12, L = 17, c_Q = 4.5, \lambda_A = 11.5$, and $\kappa \sim$ Uniform$[1.4, 1.5]$), all types remain unclipped and a^{SE} rises from 0.211 to 0.230, confirming the necessity of the condition.*

In the benchmark calibration ($k = 10, L = 10, c_Q = 2, b = 1.2, \kappa \sim$ Uniform$[0.2, 0.6]$), unclipped moments give $\bar{R}_0 = -3.73$ and $\alpha_2 = +2.47$. On the physically admissible domain for binary attributes ($g \in [0.50, 0.65]$), followers are subject to boundary clipping $m^*(\kappa) = \min\{k, \dots\}$. Evaluating the exact clipped model via numerical quadrature (`underspec_sim/verifications/verify_prop5_guessing_trap.py`): as guessing capability improves from $g = 0.50$ to $g = 0.60$, the equilibrium ask rate collapses from $a^{SE} = 0.661$ to $a^{SE} = 0.424$. In response to this withdrawal of clarification, average prompt specification expands from $\mathbb{E}[m] = 7.728$ to 7.953, and expected user welfare declines from $W = 82.134$ to 81.843. Across random parameter sweeps in the clipped model, simultaneous prompt lengthening and welfare reduction occur in approximately 7% of overall draws, and in 35% of draws with an interior a^{SE}.

This property proves essential for the comparative analysis in Section 7: Model I exhibits welfare loss relative to the first best primarily through objective bias, which shifts a^{SE} (Corollary 2) and suppresses the operational clarification rate.

6 Model II: A Screening (Menu) Policy

Now let the leader condition its ask rate on the user's declared specification level (equivalently, by the revelation principle, offer a menu of bundles that users self-select into). For tractability and clean analytical exposition, we restrict attention to two types $\kappa_L < \kappa_H$ with population shares f_L, f_H (the continuum-type extension using the standard Mirrlees technique is discussed in Section 10). Let the leader choose bundles $(m_L, a_L), (m_H, a_H)$ subject to

$$\text{IC}_\kappa: \;\; U(m_\kappa, a_\kappa; \kappa) \geq U(m_{\kappa'}, a_{\kappa'}; \kappa), \qquad \text{IR}_\kappa: \;\; U(m_\kappa, a_\kappa; \kappa) \geq \underline{U},$$

to maximize $f_L \Pi_{\kappa_L}(m_L, a_L) + f_H \Pi_{\kappa_H}(m_H, a_H)$. We normalize the follower's reservation utility to $\underline{U} = 0$, representing complete task abandonment without AI assistance. In practice, users also possess the endogenous outside option of proceeding with a non-asking baseline assistant ($a = 0$), which yields reservation utility $\underline{U}(\kappa) \equiv \max_m U(m, 0; \kappa)$. For low-cost types ($\kappa \leq \Lambda/k$), this baseline yields $\underline{U}(\kappa) = V - \frac{\kappa}{2}k^2$. Under the optimal screening menu derived below, type L receives the bundle $(k, 0)$, yielding $U(k, 0; \kappa_L) = V - \frac{\kappa_L}{2}k^2 \equiv \underline{U}(\kappa_L)$. Under this endogenous outside option, individual rationality binds exactly at the bottom of the type distribution (IR_L holds with equality), while type H's rent $U_H - \underline{U}(\kappa_H) > 0$ remains strictly positive. This provides a natural counterpart to classical regulatory screening (such as Baron–Myerson), where the least-costly agent is driven to its participation boundary.

6.1 Single-crossing

The marginal rate of substitution between m and a along an indifference curve in (m, a)-space is given by the slope $\text{MRS}(m, a; \kappa) \equiv -\frac{\partial U/\partial m}{\partial U/\partial a}$. From user utility $U(m, a; \kappa) = V - \Lambda(1 - a)(k - m) - \frac{\kappa}{2}m^2 - ac_Q(k - m)$, we have $\partial U/\partial m = \Lambda(1 - a) + ac_Q - \kappa m$ and $\partial U/\partial a = s(k - m)$, where

$s \equiv \Lambda - c_Q > 0$ is the net social gain from asking. Hence,

$$\mathrm{MRS}(m, a; \kappa) \;=\; \frac{\kappa m - [\Lambda - as]}{s(k-m)}, \qquad \frac{\partial \mathrm{MRS}}{\partial \kappa} \;=\; \frac{m}{s(k-m)} \;>\; 0,$$

which holds strictly for all interior specification levels $0 < m < k$ under the maintained condition $s > 0$. Higher-κ types therefore have strictly steeper indifference curves in (m, a)-space: they require more compensating clarification service a per unit increase in specification effort m. This satisfies the standard Spence–Mirrlees single-crossing property (Spence, 1973) and underwrites the screening analysis below.

6.2 No distortion without bias

Proposition 6 (Efficiency survives private information when the leader is unbiased)**.** *If $\mu_A = 1$ and $\lambda_A = c_Q$ (so $\Pi_\kappa \equiv U(\cdot; \kappa)$ for every κ), then implementing each type's own first-best bundle, $(m_\kappa, a_\kappa) = (m^{FB}(\kappa), a^{FB}(\kappa))$ from Proposition 3, is incentive compatible, and no further mechanism design can do better for the leader.*

Proof. $(m^{FB}(\kappa), a^{FB}(\kappa))$ maximizes $U(\cdot; \kappa)$ by definition, so for any $\kappa' \neq \kappa$,

$$U(m^{FB}(\kappa), a^{FB}(\kappa); \kappa) \;\geq\; U(m^{FB}(\kappa'), a^{FB}(\kappa'); \kappa)$$

holds trivially: no bundle can give type κ more utility than its own maximizer. IC holds with no distortion needed. Since $\Pi_\kappa \equiv U(\cdot; \kappa)$, the leader's payoff from serving each true type is maximized at that type's own first best, so this menu is also leader-optimal. □

This result highlights a fundamental distinction between human–AI coordination and classical adverse selection (such as Baron–Myerson or Laffont–Tirole). In regulatory and auction models, the principal pays monetary transfers to extract surplus and thus treats informational rents as a pure loss. Here, by contrast, the leader's objective can coincide directly with user welfare; when it does, incentive compatibility is costless, and private information incurs no welfare penalty. Distortion in this setting is therefore a pure signature of objective misalignment, not of asymmetric information per se.

Remark 7 (Degeneracy of the Unbiased Menu on the Asking Margin)**.** *Under Proposition 3, $a^{FB}(\kappa) = 1$ for every type in the population on the physically feasible domain $m \leq k$. The unbiased first-best menu is therefore $(m^{FB}(\kappa), 1) = (\min\{k, c_Q/\kappa\}, 1)$, which offers uniform asking $a = 1$ across all types and leaves followers to freely choose their preferred specification effort. This menu does not screen across the asking margin: scalar pooling at $a^{SE} = 1$ (Corollary 4) implements each type's first best identically. Consequently, under linear risk on the feasible domain, the welfare gap between pooling and screening at zero bias is identically zero ($\Pi_{II} - \Pi_I = 0$). A positive instrument-restriction gap at zero bias emerges only in settings where first best itself prescribes different asking policies across types, such as the exact conjunctive success model (Section 10) or under attribute loss heterogeneity.*

6.3 Distortion under bias

When $\Pi_\kappa \neq U(\cdot; \kappa)$ (bias present), the leader's preferred bundle per type,

$$(m^B_\kappa, a^B_\kappa) \;=\; \arg\max_{m,a} \Pi_\kappa(m, a)$$

(computable in closed form by the same method as Proposition 3, with Λ, c_Q replaced by their μ_A, λ_A-weighted counterparts), no longer automatically satisfies IC, because Π_κ and the user's true $U(\cdot;\kappa)$ now rank bundles differently.

Proposition 7 (Second-Best Screening Menu and Distortion Regimes)**.** *Assume $\Lambda > c_Q$, under-asking bias ($0 < \lambda_A - c_Q < \mu_A(\Lambda - c_Q)$, so that $c_Q < c_Q^{eff} < \Lambda$), and cost heterogeneity $\kappa_L < c_Q^{eff}/k < \kappa_H$. Let population shares be $f_L, f_H > 0$ with $f_L + f_H = 1$, define the relative share weight $t \equiv \frac{f_H}{f_L}\frac{s-\tilde{\rho}}{s}$, and define the stationary candidate:*

$$m_H^* \;\equiv\; \frac{\Lambda\tilde{\rho}}{s(\kappa_H - \kappa_L) + \tilde{\rho}\kappa_L}.$$

Define the complexity thresholds $k_{slack} \equiv \frac{2c_Q}{\kappa_L} - \frac{c_Q^{eff}}{\kappa_H}$ and $k_{crit} \equiv \frac{2c_Q}{\kappa_L} - m_H^*$, where $k_{slack} < k_{crit}$ if and only if $m_H^* < m_H^B$. Then:

1. ***Unconstrained slack regime ($k \le k_{slack}$):*** *The unconstrained menu $((k,0),(m_H^B,1))$ satisfies IC_L with strict slack ($\kappa_L(k + m_H^B) \le 2c_Q$). Both types receive their unconstrained preferred bundles, with zero distortion in effort and asking.*

2. ***Effort-distortion regime ($k_{slack} < k \le k_{crit}$, non-empty when $m_H^* < m_H^B$):*** *IC_L binds, but asking remains at the efficient level $a_H = 1$. The distortion manifests purely in downward effort distortion on the high-cost type:*

$$m_H^{SB} \;=\; \frac{2c_Q}{\kappa_L} - k \;<\; m_H^B, \qquad a_H^{SB} \;=\; 1.$$

   When $m_H^* \ge m_H^B$, $k_{slack} \ge k_{crit}$, and this intermediate effort-distortion regime is empty.

3. ***Asking-distortion regime ($k > \max\{k_{slack}, k_{crit}\}$ with $k \ge m_H^*$):*** *In the saturation regime $\kappa_L \le \bar{\kappa}_L \equiv \frac{c_Q^{eff} + tc_Q}{k(1+t)}$, the leader sets $(m_L^{SB}, a_L^{SB}) = (k, 0)$, and the high-cost contract is uniquely given in closed form by:*

$$m_H^{SB} \;=\; m_H^*, \qquad a_H^{SB} \;=\; 1 \;-\; \frac{\frac{\kappa_L(k+m_H^*)}{2} - c_Q}{s} \;<\; 1.$$

   Downward distortion $1 - a_H^{SB}$ intensifies as κ_L increases toward $\bar{\kappa}_L$. At $m_L = k$, zero residual attributes remain ($k - m_L = 0$), so a_L is payoff-irrelevant to both parties.

4. ***Boundary saturation regime ($k < m_H^*$):*** *When cost heterogeneity is narrow ($\kappa_H - \kappa_L$ small) so that $m_H^* > k$, the physical feasibility constraint $m_H \le k$ binds. By strict concavity of Π_{κ_H}, optimal effort hits $m_H = k$, at which point residual attributes vanish ($k - m_H = 0$), asking is payoff-irrelevant, and the menu collapses to boundary pooling at $(k, 0)$.*

5. ***Boundary relaxation:*** *If $\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$, the leader optimally offers type L clarification service ($a_L^{SB} = 1, m_L^{SB} < k$) to increase U_L, which relaxes IC_L and moderates the downward distortion on type H relative to the saturation formula (dampening downward distortion growth without reversing its comparative static).*

Proof. We establish the result in five analytical steps:

Step 1: Unconstrained biased optima. The leader objective per type is $\Pi_\kappa(m,a) = \mu_A[V - \Lambda(1-a)(k-m) - \frac{\kappa}{2}m^2 - ac_Q^{eff}(k-m)]$, where $c_Q^{eff} \equiv c_Q + (\lambda_A - c_Q)/\mu_A$. Under the maintained condition $0 < \lambda_A - c_Q < \mu_A(\Lambda - c_Q)$, effective asking friction satisfies $c_Q < c_Q^{eff} < \Lambda$. For type H with $\kappa_H > c_Q^{eff}/k$, because $c_Q^{eff} < \Lambda$, Π_{κ_H} is strictly increasing in a for any $m < k$ (slope $\mu_A(\Lambda - c_Q^{eff})(k-m) > 0$). The leader's unconstrained optimum for type H therefore sets $a_H^B = 1$ and interior specification $m_H^B = c_Q^{eff}/\kappa_H < k$. For type L with $\kappa_L < c_Q^{eff}/k$, the partial derivative with respect to effort satisfies $\partial\Pi_{\kappa_L}/\partial m = \mu_A[\Lambda(1-a) + ac_Q^{eff} - \kappa_L m] \geq \mu_A[c_Q^{eff} - \kappa_L k] > 0$ for all $m \in [0,k]$ and all $a \in [0,1]$. Hence Π_{κ_L} is strictly increasing in m, and attains its unconstrained maximum on $[0,k]\times[0,1]$ at $m_L^B = k$. At $m_L^B = k$, zero attributes remain unspecified ($k - m_L^B = 0$), so asking is payoff-irrelevant to both the user and the leader ($\Pi_{\kappa_L}(k,a) = \mu_A[V - \frac{\kappa_L}{2}k^2]$ for all a). Setting $a_L^B = 0$ is the unconstrained benchmark choice because it gives zero asking service, minimizing the temptation of type H to mimic type L.

Step 2: Upward temptation to mimic (IC_L violation). Evaluating type κ_L's payoff from mimicking the high type's unconstrained bundle $(m_H^B, 1)$ versus accepting its own bundle $(k,0)$:

$$U(m_H^B, 1; \kappa_L) - U(k, 0; \kappa_L) \;=\; (k - m_H^B)\left[\frac{\kappa_L(k + m_H^B)}{2} - c_Q\right].$$

Because $m_H^B < k$, this difference is strictly positive if and only if $\kappa_L(k + m_H^B) > 2c_Q$, which is equivalent to $k > k_{\text{slack}} \equiv \frac{2c_Q}{\kappa_L} - m_H^B = \frac{2c_Q}{\kappa_L} - \frac{c_Q^{eff}}{\kappa_H}$. When $k \leq k_{\text{slack}}$, IC_L is strictly satisfied at the unconstrained bundles, establishing claim 1.

Step 3: Relaxed problem and slackness of dropped constraints (IC_H, IR_H). We formulate the relaxed screening problem by maximizing the expected leader objective subject to IC_L alone, dropping IC_H, IR_L, and IR_H. We then establish that all dropped constraints are strictly satisfied at the relaxed optimum:

1. *Slackness of IC_H via single-crossing:* By the Spence–Mirrlees sorting condition $\frac{\partial^2 U}{\partial m\,\partial \kappa} = -m < 0$, the rent identity for any menu $((m_L, a_L), (m_H, a_H))$ is:

   $$U(m_H, a_H; \kappa_H) - U(m_L, a_L; \kappa_H) \;=\; [U(m_H, a_H; \kappa_L) - U(m_L, a_L; \kappa_L)] \;+\; \frac{\kappa_H - \kappa_L}{2}\big(m_L^2 - m_H^2\big).$$

   When $(m_L, a_L) = (k, 0)$ and IC_L binds with equality ($U(m_H, a_H; \kappa_L) = U(k, 0; \kappa_L)$), this simplifies directly to:

   $$U(m_H, a_H; \kappa_H) - U(k, 0; \kappa_H) \;=\; \frac{\kappa_H - \kappa_L}{2}\big(k^2 - m_H^2\big) \;>\; 0,$$

   which is strictly positive for any $m_H < k$ since $\kappa_H > \kappa_L$. Hence IC_H holds with strict inequality (strict slack).

2. *Slackness of IR_H and IR_L:* The user's reservation utility is their unassisted payoff $\underline{U}(\kappa) \equiv \max_{m\in[0,k]} U(m, 0; \kappa) = V - \frac{\kappa}{2}k^2 = U(k, 0; \kappa)$. For type L, $U(m_L, a_L; \kappa_L) = U(k, 0; \kappa_L) = \underline{U}(\kappa_L)$, so IR_L holds. For type H, strict slackness of IC_H immediately implies $U(m_H, a_H; \kappa_H) > U(k, 0; \kappa_H) = \underline{U}(\kappa_H)$, establishing that participation IR_H is strictly slack.

Step 4: Global optimization along the IC_L boundary and regimes. Because Π_{κ_H} is strictly increasing in a_H on $c_Q^{eff} < \Lambda$, the leader pushes a_H as high as incentive compatibility permits, so IC_L must bind at the relaxed optimum. Along the binding boundary $U(m_H, a_H; \kappa_L) = U_L^0 \equiv V - \frac{\kappa_L}{2}k^2$, utility is strictly linear in a_H, uniquely determining:

$$a_H(m_H) \;=\; 1 \;-\; \frac{U(m_H, 1; \kappa_L) - U_L^0}{s(k - m_H)} \;=\; 1 \;-\; \frac{\frac{\kappa_L(k+m_H)}{2} - c_Q}{s}.$$

Substituting this boundary into the leader's objective $\Pi_{\kappa_H}(m_H, a_H(m_H))$ yields a strictly concave quadratic in m_H:

$$\frac{d^2\Pi_{\kappa_H}}{dm_H^2} \;=\; -\,\frac{\mu_A\big[s(\kappa_H - \kappa_L) + \tilde{\rho}\kappa_L\big]}{s} \;<\; 0,$$

which attains its unique unconstrained stationary point at:

$$m_H^* \;=\; \frac{b\Lambda}{\mu_A s(\kappa_H - \kappa_L) + b\kappa_L} \;=\; \frac{\Lambda\tilde{\rho}}{s(\kappa_H - \kappa_L) + \tilde{\rho}\kappa_L}.$$

Now consider the interaction between the constraints $m_H \le k$ and $a_H \le 1$:

- If $k_{\text{slack}} < k \le k_{\text{crit}} \equiv \frac{2c_Q}{\kappa_L} - m_H^*$ (which is non-empty whenever $m_H^* < m_H^B$), then $\frac{\kappa_L(k+m_H^*)}{2} \le c_Q$. Evaluating $a_H(m_H^*)$ would yield $a_H \ge 1$. Since $a_H \le 1$, the upper bound $a_H = 1$ binds. Along $a_H = 1$, satisfying IC_L with equality requires $(k - m_H)[\frac{\kappa_L(k+m_H)}{2} - c_Q] = 0$, which uniquely sets $m_H = \frac{2c_Q}{\kappa_L} - k < m_H^B$. The distortion manifests entirely through prompt effort, establishing claim 2. When $m_H^* \ge m_H^B$, $k_{\text{slack}} \ge k_{\text{crit}}$ and this intermediate effort-distortion regime is empty.

- If $k > \max\{k_{\text{slack}}, k_{\text{crit}}\}$ and $k \ge m_H^*$, then $\frac{\kappa_L(k+m_H^*)}{2} > c_Q$, yielding interior asking distortion $a_H^{SB} = a_H(m_H^*) < 1$ with effort $m_H^{SB} = m_H^*$, establishing claim 3.

- If $k < m_H^*$ (which occurs when cost heterogeneity $\kappa_H - \kappa_L$ is small), the stationary point exceeds the physical attribute count ($m_H^* > k$). By strict concavity of $\Pi_{\kappa_H}(m_H)$ on $[0, k]$, the constrained maximizer is $m_H^{SB} = k$. At $m_H = k$, residual attributes vanish ($k - m_H = 0$), so asking is payoff-irrelevant and the menu collapses to boundary pooling at $(k, 0)$, establishing claim 4.

We now analytically determine when $(m_L^{SB}, a_L^{SB}) = (k, 0)$ is the global optimum. Consider departing from $m_L = k$ by offering type L clarification service ($a_L = 1$) with $m_L < k$. Since $\kappa_L > c_Q/k$, reducing m_L below k increases type L's utility at marginal rate $\frac{\partial U}{\partial m_L}\Big|_{m_L=k} = c_Q - \kappa_L k < 0$, yielding a utility increase of $\kappa_L k - c_Q > 0$. With population shares f_L, f_H, increasing type L's utility by $+1$ relaxes IC_L, allowing a_H to increase by $\frac{1}{s(k-m_H)}$, which increases the leader's payoff on type H at rate $f_H\frac{\mu_A s - b}{s}(\kappa_L k - c_Q)$. Conversely, granting type L clarification costs the leader $f_L\mu_A(c_Q^{eff} - \kappa_L k)$ per unit of reduction in m_L. Departing from $(k, 0)$ is strictly unprofitable if and only if the marginal gain on type H is weakly less than the marginal cost on type L:

$$f_H\frac{\mu_A s - b}{s}(\kappa_L k - c_Q) \;\le\; f_L\mu_A(c_Q^{eff} - \kappa_L k).$$

Dividing by $f_L\mu_A$ and substituting $\tilde{\rho} = b/\mu_A$ and $t \equiv \frac{f_H}{f_L}\frac{s-\tilde{\rho}}{s}$, this simplifies to:

$$t(\kappa_L k - c_Q) \;\le\; c_Q^{eff} - \kappa_L k \iff \kappa_L \;\le\; \bar{\kappa}_L \;\equiv\; \frac{c_Q^{eff} + tc_Q}{k(1+t)}.$$

When $f_L = f_H = 0.5$, $t = \frac{s-\tilde{\rho}}{s}$ and $\bar{\kappa}_L$ simplifies to $\frac{sc_Q^{eff}+(s-\tilde{\rho})c_Q}{k(2s-\tilde{\rho})}$. When $\kappa_L \leq \bar{\kappa}_L$, the leader strictly prefers to keep type L at the corner $(k, 0)$. In this regime, $(m_L^{SB}, a_L^{SB}) = (k, 0)$ is the exact global second-best optimum. When $\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$, the leader departs from $(k, 0)$ by setting $a_L^{SB} = 1$ and $m_L^{SB} < k$. Raising U_L relaxes IC_L, moderating the downward distortion on type H relative to the saturation formula, establishing claim 5.

Step 5: Comparative statics across the cost spectrum. In the low-cost saturation regime ($\kappa_L \leq \bar{\kappa}_L$ with $k > k_{\text{crit}}$ and $k \geq m_H^*$):

$$1 - a_H^{SB} \;=\; \frac{\frac{\kappa_L(k+m_H^*)}{2} - c_Q}{s}.$$

As κ_L increases toward $\bar{\kappa}_L$, type L's utility from completing k attributes decreases, intensifying the temptation to mimic the asking bundle. To deter mimicry, the leader must distort a_H^{SB} progressively downward (lower a_H). In the boundary relaxation regime ($\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$), the leader relieves this pressure by granting clarification to type L ($a_L = 1, m_L < k$), which raises U_L and moderates the downward distortion on type H relative to the saturation formula, dampening distortion growth without reversing its negative direction. Finally, when friction reaches or exceeds task stakes ($c_Q^{eff} \geq \Lambda$, or $\lambda_A - c_Q \geq \mu_A(\Lambda - c_Q)$), the leader unconstrainedly prefers zero asking for all types ($\partial\Pi/\partial a \leq 0$). Clarification is uniformly shut down: $a_L = a_H = 0$ and $m_L = m_H = k$. Both types receive the identical bundle $(k, 0)$, so IC_L and IC_H hold with equality ($0 = 0$). This boundary co-activity reflects technological clarification shutdown due to prohibitive friction, whereas genuine adverse-selection screening distortion ($a_H^{SB} < a_H^B = 1$) operates strictly in the cost-effective asking regime $c_Q < c_Q^{eff} < \Lambda$. □

Remark 8 (Active Constraint Regimes and Effort versus Asking Distortions)**.** *In classical transfer-based screening (such as Baron–Myerson), IC_L and IR_H typically bind while IC_H and IR_L remain slack. In our setting, solving the full four-variable constrained problem reveals five distinct operational regimes:*

1. Unconstrained slack regime ($k \leq k_{\text{slack}}$): *When task complexity is small, the temptation of type L to mimic is absent ($\kappa_L(k+m_H^B) \leq 2c_Q$), so IC_L and IC_H are strictly slack at the unconstrained menu.*

2. Effort-distortion regime ($k_{\text{slack}} < k \leq k_{\text{crit}}$): *Type L faces a temptation to mimic, but the assistant absorbs the entire distortion through specification effort ($m_H = \frac{2c_Q}{\kappa_L} - k < m_H^B$) while preserving full clarification service ($a_H = 1$). Here IC_L binds while asking is undistorted.*

3. Adverse-selection asking distortion regime ($k > k_{\text{crit}}$ with $k \geq m_H^*$): *Specification distortion alone is insufficient, forcing the assistant to distort asking downward ($a_H^{SB} < 1$) alongside effort $m_H = m_H^*$.*

4. Boundary saturation regime ($k < m_H^*$): *Narrow cost heterogeneity forces the interior effort candidate above task complexity ($m_H^* > k$), causing the physical ceiling $m_H \leq k$ to bind and collapsing the menu to boundary pooling at $(k, 0)$.*

5. Prohibitive friction regime ($c_Q^{eff} \geq \Lambda$, equivalently $\lambda_A - c_Q \geq \mu_A(\Lambda - c_Q)$): *Asking friction exceeds the entire task risk resolved, so the leader unconstrainedly prefers zero asking for all types ($\partial\Pi/\partial a \leq 0$). Clarification is uniformly shut down: $a_L = a_H = 0$ and $m_L = m_H = k$.*

Both types receive the identical bundle $(k, 0)$, so IC_L and IC_H hold with equality trivially. This boundary co-activity reflects technological shutdown, not an adverse-selection screening distortion.

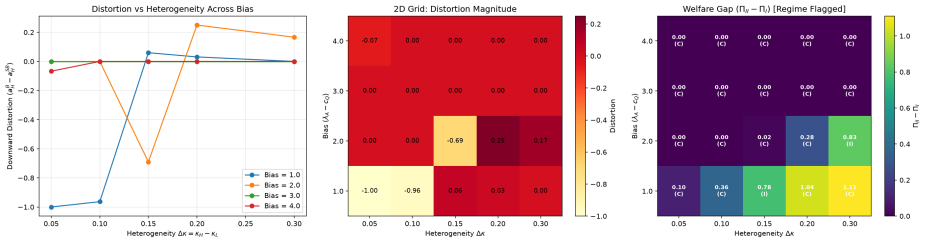


Figure 4: Screening menu distortion across cost heterogeneity (κ_L, κ_H) and provider friction λ_A, illustrating adverse-selection downward distortion $a_H^{SB} < 1$.

Remark 9 (Directionality of Policy Misalignment)**.** *The downward distortion in Proposition 7 is specific to the under-asking bias case ($0 < \lambda_A - c_Q < \mu_A(\Lambda - c_Q)$), which reflects the empirical reality of turn-penalizing satisfaction metrics. Under an over-asking bias ($\lambda_A < c_Q$), the direction of distortion can reverse, as the leader would favor excessive clarification.*

7 Comparing the Two Regimes

Corollary 5 (Menus weakly dominate pooling)**.** *The leader's optimal expected payoff under Model II is weakly greater than under Model I: any pooling policy a is a feasible (degenerate) menu with $a_L = a_H = a$, trivially satisfying IC, so the unconstrained menu optimum is at least as good.*

Beyond payoff dominance, the central comparison lies in the structural decomposition of welfare losses across the two regimes, enabled by Propositions 4–7:

- **Model I (pooling):** Under the linear-risk specification on the physically feasible domain $m \in [0, k]$, an unbiased assistant selects $a^{SE} = 1$, which implements the first-best policy $a^{FB}(\kappa) = 1$ for every type in the population (Proposition 3). Consequently, in the absence of bias ($\lambda_A = c_Q$), the pooling restriction incurs zero welfare loss ($\Pi_{II} - \Pi_I = 0$). Welfare loss under pooling arises when objective bias is present ($\lambda_A > c_Q$), which suppresses the single ask rate a^{SE} below the socially efficient level (Corollary 2), or in the exact conjunctive model (Section 10), where first-best policies differ across types ($a^{FB} \in \{0, 1\}$) and a uniform rate cannot simultaneously tailor to heterogeneous needs.
- **Model II (screening)** incurs welfare loss from bias *alone* (Proposition 6): in the absence of bias, screening recovers full efficiency despite private cost information. Under bias, the loss takes the form of the downward distortion characterized in Proposition 7. This represents a fundamentally different mechanism from the instrument-restriction loss of pooling: it stems not from an inability to condition on observed effort, but from the requirement to prevent opportunistic mimicking across types.

7.1 Sensitivity of the Unbiased Pooling Corner and Physical Feasibility

While the qualitative finding of Corollary 4, that unbiased pooling is always a corner solution rather than an interior compromise, holds universally across the parameter space, the unclipped

corner difference formula $\Pi(1) - \Pi(0) = (\Lambda - c_Q)[k - k^*]$ relies on unconstrained best responses. Table 1 documents this unconstrained boundary across varying task sizes k and baseline guessing accuracies g, holding $L = 10, c_Q = 2$, and $\kappa \sim \text{Uniform}[0.2, 0.6]$ (where $\mathbb{E}[1/\kappa] \approx 2.747$).

g	$\Lambda = L(1-g)$	Threshold k^*	Task Size k	Unclipped $\Delta\Pi$	Clipped $\Delta\Pi$
0.30	7.0	12.36	5	−36.80	+0.15
0.30	7.0	12.36	10	−11.80	+5.49
0.30	7.0	12.36	15	+13.20	+20.10
0.50	5.0	9.61	5	−13.84	+0.15
0.50	5.0	9.61	9	−1.84	+3.69
0.50	5.0	9.61	10 (Benchmark)	+1.16	+5.42
0.50	5.0	9.61	12	+7.16	+9.62
0.50	5.0	9.61	15	+16.16	+17.12
0.70	3.0	6.87	5	−1.87	+0.15
0.70	3.0	6.87	8	+1.13	+1.71
0.70	3.0	6.87	10	+3.13	+3.32

Table 1: Sensitivity of the unbiased pooling corner choice across task complexity k and guessing capability g, for $L = 10, c_Q = 2, \kappa \sim \text{Uniform}[0.2, 0.6]$, where $\Delta\Pi \equiv \Pi(1) - \Pi(0)$. Columns compare the unconstrained closed form $(\Lambda - c_Q)[k - k^*]$ against the physically feasible clipped payoff difference computed via vectorized numerical integration (`underspec_sim/verifications/generate_table1.py`, recorded in `outputs/table1_unbiased_sensitivity.csv`). In the unclipped algebra, guessing appears to win whenever $k < k^*$ (e.g., −36.80 at $g = 0.30, k = 5$), but when followers are constrained to feasible effort $m \leq k$, asking strictly dominates across all rows (+0.15 at $g = 0.30, k = 5$), confirming Proposition 3. While binary symmetric attributes naturally correspond to baseline guessing accuracy $g = 0.50$, configurations with $g = 0.30$ and $g = 0.70$ represent tasks with non-binary attribute domains (such as multi-choice syntactical options where random guessing yields $1/M < 0.5$, or informed priors where guessing yields > 0.5).

As shown in Table 1, once followers are restricted to feasible specification effort $m \in [0, k]$, asking ($a^{SE} = 1$) weakly dominates guessing across all 11 configurations under linear risk. The apparent dominance of guessing in the unclipped column stems entirely from low-cost types choosing unfeasible specification levels $m = \Lambda/\kappa > k$. At $k = 5$, for all $\kappa \in [0.2, 0.6]$, the unassisted specification choice under guessing is $\min\{5, \Lambda/\kappa\} = 5 = k$ across all $g \in \{0.30, 0.50, 0.70\}$, because $\Lambda \geq 3.0$ and $\kappa \leq 0.6$. When followers fully saturate specification effort under silence, the task risk parameter Λ drops out completely from the comparison, and the net gain from asking $\Delta\Pi = \mathbb{E}[U(m^*(1); 1) - U(k; 0)] = +0.15$ depends strictly on $c_Q = 2$ and κ. In contrast, when evaluated under the exact conjunctive success function $q(a, g)^{k-m}$, non-linear probability curvature generates genuine interior trade-offs where guessing beats asking even on the interior domain (Section 10).

Remark 10 (Welfare vs Leader Payoff Ordering Across Bias Bands)**.** *While screening weakly dominates pooling in leader payoff ($\Pi_{II} \geq \Pi_I$) by revealed preference (Corollary 5), the ranking of user welfare $W \equiv \mathbb{E}[U]$ depends non-monotonically on provider asking friction λ_A (evaluated at $\kappa_L = 0.3, \kappa_H = 0.5, f_L = 0.5, k = 10, L = 10, g = 0.5, c_Q = 2, V = 100$, and $\mu_A = 1.0$, verified in `underspec_sim/verifications/verify_welfare_ordering.py`):*

1. Mild bias ($\lambda_A \in [2.5, 3.0]$): *Pooling maintains universal clarification ($a^{SE} = 1$) and achieves strictly higher user welfare than screening ($W_{pool} = 85.33$ vs $W_{screen} = 85.00$ at $\lambda_A = 2.5$, gap -0.33; $W_{pool} = 85.33$ vs $W_{screen} = 84.52$ at $\lambda_A = 3.0$, gap -0.81). Importantly, the welfare loss*

under screening at $\lambda_A = 2.5$ arises entirely through effort dictation*: because $k < k_{slack} = 10.33$, asking is undistorted ($a_L = a_H = 1.0$), but the biased assistant demands excessive upstream specification ($m_L = 8.33 > 6.67, m_H = 5.00 > 4.00$) to economize on provider query costs. At $\lambda_A = 3.0$, task complexity exceeds the asking threshold ($k > k_{crit} = 8.33$), so downward asking distortion sets in ($a_H^{SB} = 0.833 < 1.0$), further depressing user welfare.*

2. Moderate bias ($\lambda_A \in [3.5, 4.5]$): *Pooling collapses to zero asking ($a^{SE} = 0$, yielding baseline unassisted welfare $W_{pool} = 80.00$), whereas screening continues to provide partial clarification to terse users ($a_H^{SB} > 0$), delivering strictly higher user welfare ($W_{screen} = 82.45$ at $\lambda_A = 3.5$, gap $+2.45$; $W_{screen} = 81.53$ at $\lambda_A = 4.0$, gap $+1.53$; $W_{screen} = 80.71$ at $\lambda_A = 4.5$, gap $+0.71$).*

3. Severe / prohibitive friction ($\lambda_A \geq 5.0$): *Clarification is priced out under both regimes ($a^{SE} = 0$ and $a_L = a_H = 0$), so user welfare ties at $W_{pool} = W_{screen} = 80.00$ (gap 0.00).*

In the exact conjunctive model, an additional structural instrument-restriction gap appears even at zero bias, separating technological restriction from behavioral misalignment.

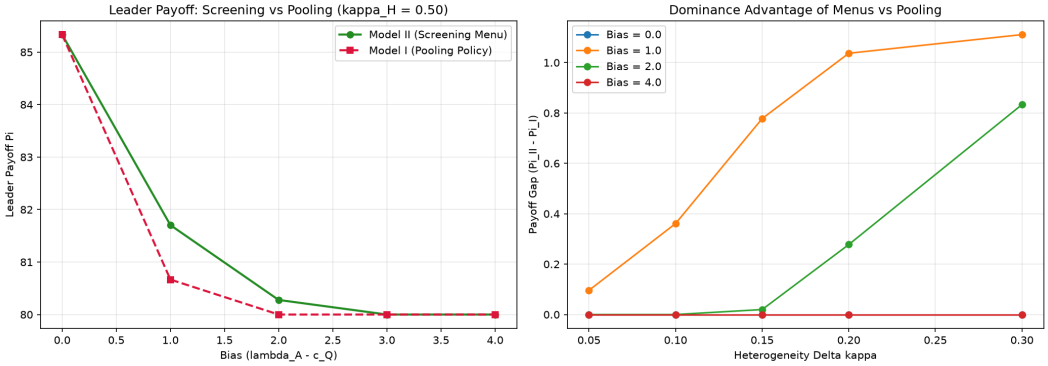


Figure 5: Leader payoff under Model II (screening menu) versus Model I (pooling policy) across assistant bias $\lambda_A - c_Q$ and type heterogeneity $\kappa_H - \kappa_L$, illustrating weak dominance ($\Pi_{II} \geq \Pi_I$) and the zero-gap baseline at zero bias under linear risk.

8 Testable Predictions

P1 (Prop. 1, Cor. 1) Holding the assistant's policy fixed, higher guessing accuracy is associated with shorter specifications alongside weakly higher user welfare, and never with an overall welfare decline despite the reduction in prompt length.

P1′ (Model I remark) Holding accuracy fixed, the effect of a higher *ask rate* on specification length flips sign depending on whether $c_Q \gtrless \Lambda = L(1-g)$: where clarifying questions are inexpensive relative to the risk they resolve, users specify less as the ask rate rises (substitution); where questions are relatively costly to answer, users specify *more* (pre-emption). This prediction operates within the model and is distinct from P1's accuracy channel.

P2 (Prop. 2) The correlation between turn-level satisfaction and eventual task success weakens or inverts specifically for under-asking assistants, and can be benchmarked against the derived a^{SE} of Proposition 4 rather than an assumed ask rate.

P3 Making ask-rate statistics visible to users shifts specification length toward the sophisticated best response analyzed in Section 3.3.

P4 (Prop. 7) Ask rates should rise with a user's revealed cost of providing detail (terseness) when clarification policies are screening menus or under nonlinear success models with attribute heterogeneity, rather than responding exclusively to model confidence.

P5 Users with lower self-reported error-catching rates exhibit declining specification effort across an interactive session (following Section 3's dynamic channel).

P6 (Cor. 4, Prop. 7) Under an assistant with asking friction bias ($\lambda_A > c_Q$), a menu-screening policy achieves weakly higher leader payoff than pooling ($\Pi_{II} \geq \Pi_I$) by tailoring specification requirements to user capabilities. For user welfare ($W \equiv \mathbb{E}[U]$), however, the ranking depends non-monotonically on friction: under mild bias ($\lambda_A \in (c_Q, \lambda_A^*)$, where $\lambda_A^* \approx 3.33$ in the calibrated baseline is the threshold where pooling asking collapses), pooling maintains universal clarification ($a^{SE} = 1$) and delivers higher user welfare than screening (which suppresses user surplus via inflated effort requirements and downward asking distortion); under severe bias ($\lambda_A \geq \lambda_A^*$), friction forces pooling to collapse to zero asking ($a^{SE} = 0$), whereas screening preserves partial clarification for high-cost users and restores higher user welfare. In settings characterized by nonlinear exact success or heterogeneous attribute stakes (where first best prescribes asking only for terse users), an unbiased pooling policy also incurs an instrument-restriction loss by forcing uniform asking.

P7 (Prop. 7) Even when an assistant can condition on declared specification effort, terse users receive fewer clarifying questions than the assistant's unconstrained per-type preference would prescribe ($a_H^{SB} < a_H^B$). This downward distortion arises specifically from the incentive-compatibility constraint IC_L. In the low-cost saturation regime ($\kappa_L \leq \bar{\kappa}_L$), downward distortion intensifies as cost heterogeneity narrows (κ_L approaches $\bar{\kappa}_L$), requiring deeper distortion to prevent mimicry; in the boundary relaxation regime ($\bar{\kappa}_L < \kappa_L < c_Q^{eff}/k$), the assistant grants clarification to low-cost users ($a_L^{SB} = 1$), relaxing IC_L and strictly dampening downward distortion on high-cost users relative to the saturated formula (preventing the steep collapse that would occur if a_L were held at 0).

P8 (Prop. 5) Under an assistant with under-asking friction bias ($\lambda_A > c_Q$), higher model guessing capability g can perversely lengthen user prompts ($d\mathbb{E}[m^*]/dg > 0$) and lower user welfare ($dW/dg < 0$). This equilibrium capability reversal occurs if and only if the assistant's ask-rate reduction is sufficiently steep to outpace direct guessing gains ($-\frac{da^{SE}}{dg} > \frac{L(1-a^{SE})}{\Lambda(g)-c_Q}$), forcing users to over-specify upstream to defensively insure against guessing errors.

9 Empirical Strategy and Experimental Design

The behavioral predictions P1–P8 (including P1′) translate our contract-theoretic mechanisms into observable margins for experimental testing. In naturalistic software engineering logs, however, assistant policy bias λ_A and user specification costs κ are typically confounded with task difficulty and user experience. To causally test our predictions, we propose a between-subjects randomized controlled trial on software developers performing standardized implementation tasks:

Experimental design and treatments. Participants are randomly assigned to interact with code assistants operating under distinct clarification regimes:

1. *Treatment A (Low-asking policy, $a = 0.1$):* The assistant aggressively guesses ambiguous attributes with minimal clarification.

2. *Treatment B (High-asking policy, $a = 0.8$):* The assistant frequently asks targeted clarifying questions before generating code.

3. *Treatment C (Menu-conditioned screening policy):* The assistant dynamically selects its clarification rate based on the structural completeness of the initial user prompt, implementing the screening menu derived in Section 6.

4. *Control (Unbiased benchmark):* The assistant's clarification rate is calibrated to the user-optimal benchmark.

Measurement, identification, and power analysis. For each programming session, instrumentation logs: (i) upstream prompt length and specified invariants/edge cases (m); (ii) clarification turn count ($a(k - m)$); (iii) downstream test suite execution and correctness; and (iv) post-session subjective developer satisfaction. By exogenously varying the clarification policy a, this protocol directly tests whether users economize on prompt specification in response to anticipated assistance ($m^*(a)$ effort substitution, Prediction P1), distinguishes the substitution channel ($\Lambda > c_Q$) from the pre-emption channel ($c_Q > \Lambda$, Prediction P1′), and evaluates whether clarification responds to user specification terseness (Prediction P4).

To identify user specification cost parameters κ without confounding from unobserved task difficulty, the protocol includes a standardized pre-trial onboarding task where participants articulate requirements without assistant interaction, establishing an individual-level baseline specification effort per unit of task complexity. Assistant asking friction λ_A is exogenously manipulated through provider system prompt instructions, conversational turn penalties, and backend latency budgets. A statistical power calculation indicates that detecting a medium effect size ($d = 0.35$ standard deviations in prompt length m between low-asking and high-asking treatments) at $\alpha = 0.05$ with 80% statistical power ($1 - \beta = 0.80$) requires approximately $N = 134$ developers per treatment arm, for a total sample of $N \approx 536$ participants across the four arms.

Ethics and human subjects considerations. The proposed human evaluation involves non-invasive coding tasks. Participating developers should be recruited through professional developer platforms, provide explicit informed consent, and receive compensation at standard market rates for professional software engineering (e.g., \$50–\$75 per hour). All logged code, prompts, and telemetry must be fully anonymized prior to analysis, with no personally identifiable information recorded. The study protocol should be submitted to the relevant Institutional Review Board (IRB) for formal exemption or expedited review under institutional guidelines for research involving benign behavioral interventions. To facilitate reproducibility and protocol validation prior to live human deployment, we provide a companion open-source simulation harness (`underspec_sim/experiments_llm`) supporting both deterministic offline dry-run evaluation with mock LLM client responses and live API interactions across pooling and screening treatments.

10 Discussion and Limitations

The linear-risk formulation is the main tractability baseline analyzed in Sections 4–7. Rather than viewing linear risk solely as a small-risk Taylor approximation, it is properly understood as an

independent, additive-defect primitive model:

$$U(m, a; \kappa) \;=\; V \;-\; L(1-q)(k-m) \;-\; \frac{\kappa}{2}m^2 \;-\; a\, c_Q\,(k-m),$$

where L represents the localized defect cost per unresolved attribute and cumulative defect loss satisfies $kL \leq V$. In this additive environment, each unaddressed attribute produces an independent defect without destroying the entire project. By contrast, the exact conjunctive specification $Vq(a,g)^{k-m}$ represents a systemic bottleneck environment where all attributes are strictly complementary and a single error destroys the entire deliverable. To calibrate the exact model against the linear baseline, note that for benchmark task size $k = 10$, total project valuation $V = 100$ corresponds precisely to cumulative defect loss $kL = 100$ (with per-attribute stake $L = 10$). To investigate the role of payoff curvature, we evaluate both the unbiased pooling optimum and the biased screening menu directly under the exact conjunctive payoff $q(a,g)^{k-m}$ via direct numerical grid search and constrained optimization (implemented in `verify_exact_conjunctive.py`, recorded in `outputs/exact_conjunctive_robustness.csv`). Rather than merely replicating the linear model, the exact conjunctive specification represents a distinct, richer economic environment where probability curvature introduces genuine interior trade-offs.

Under this exact nonlinear specification across task complexity $k \in [3, 20]$ and guessing accuracy $g \in [0.50, 0.95]$ (30 grid points):

1. The corner-solution property of Corollary 4 persists universally ($a^{SE}_{\text{exact}} \in \{0, 1\}$ in all 30 configurations).

2. Unbiased pooling choices agree between the exact model and the linear Taylor-expansion baseline ($L = V = 100$) in 25 of 30 configurations (83.3%), with all 5 divergences concentrated exclusively at small $k = 3$ where compounding (q^3) makes autonomous guessing preferred over question friction. For all $k \geq 5$ across all $g \in [0.50, 0.95]$, both models select universal clarification $a^{SE} = 1.0$. Under the localized additive-defect baseline ($L = 10$), agreement is 17 of 30 configurations (56.7%), with divergences at high $g \geq 0.90$ where localized attribute stakes make guessing cost-effective earlier than systemic failure stakes. This nonlinear curvature creates a positive instrument-restriction loss at zero bias in the exact model that is absent in linear risk.

3. Downward screening distortion ($a^{SB}_H < a^B_H$) holds in 20 of 30 cases (67% overall: 20% at $k \in \{3, 5\}$, 80% at $k \in \{8, 20\}$, and 100% at $k \in \{10, 15\}$). The low incidence of distortion at small k stems from boundary saturation: when tasks are simple ($k \leq 5$), users can fully specify all attributes ($m = k$), leaving no residual ambiguity for clarification to resolve ($k - m = 0$). Consequently, both IC_L and IC_H are active in 15 of 30 cases, concentrated entirely at small task sizes ($k \leq 8$).

Furthermore, numerical solutions of the full four-variable constrained menu problem under the exact payoff across $\lambda_A \in [c_Q, 10c_Q]$ and $\mu_A \in [0.1, 1.0]$ (`verify_exact_bias_sweep.py`, recorded in `outputs/exact_bias_sweep.csv` and Figure 6) illustrate the boundary transition to clarification shutdown. In the linear model with $\mu_A = 1.0$, asking friction reaches or exceeds task stakes ($c^{eff}_Q \geq \Lambda$) at $\lambda_A \geq 5.0$. Beyond this threshold, the leader unconstrainedly prefers zero asking for all types ($a^B_L = a^B_H = 0$). Under the exact nonlinear payoff, this technological clarification shutdown threshold shifts outward to $\lambda_A \approx 8.0$. In this shutdown regime, both types choose full specification $m = k$ and receive $(k, 0)$, so IC_H and IC_L both hold with equality ($0 = 0$). This co-activity is not an adverse-selection screening distortion, but a technological shutdown artifact where clarification is universally priced out by severe asking friction.

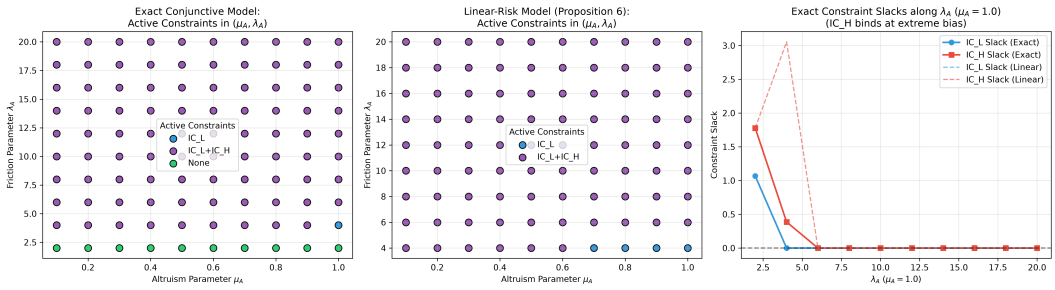


Figure 6: Screening menu allocations and active constraint boundaries under the exact conjunctive payoff across assistant asking friction λ_A and weight μ_A, illustrating the transition to clarification shutdown at $\lambda_A \approx 8.0$.

Another important limitation concerns the off-menu contract schedule. Proposition 7 utilizes the standard direct revelation principle, implementing the second-best menu via a discontinuous off-menu policy $a(m) = 0$ for all $m \neq m_H^*$. While theoretically canonical, implementing discontinuous punishment schedules in conversational AI systems may be practically brittle or unintuitive for end users. Exploring smooth, continuous UI policies $a(m)$ or graded clarification schedules that remain robust to prompt gaming offers a fruitful avenue for applied mechanism design.

Regarding the regularity condition (Assumption 1), the global Topkis submodularity condition requires $\partial^2 U/\partial m \partial g \leq 0$ across the entire domain $m \in [0, k]$, which reduces to $k \ln q \geq -1$. In a broad numerical parameter grid, this global condition holds in 24.8% of configurations. However, for monotone comparative statics of the optimal choice $m^*(g)$, decreasing differences need only hold locally at the endogenous optimum m^*. The corresponding local condition $(k - m^*) \ln q \geq -1$ holds in 89.9% of the parameter grid (`verify_assumption1.py`). Furthermore, even in the remaining 10.1% of cases where the cross-partial turns positive, prompt specification $m^*(g)$ remains weakly decreasing in 100% of cases due to boundary clipping at $m \in \{0, k\}$ (`verify_monotonicity_survival.py`, recorded in `outputs/monotonicity_survival.csv`). Thus, Topkis submodularity is a sufficient rather than necessary condition for monotonic effort reduction.

A stylized modeling assumption is attribute symmetry (k binary attributes with identical loss L and guessing probability g). In practical software engineering, attributes exhibit substantial loss heterogeneity: omitting an architectural invariant or security convention causes severe failure ($L_i \gg 0$), while omitting a stylistic naming convention carries negligible cost ($L_j \approx 0$). In an extension with attribute-specific losses L_i, the user's objective becomes $U = V \prod_i q_i - c(m) - \sum_i a_i c_Q$. Ordering attributes by descending risk stake $\Lambda_i = L_i(1 - g_i)$, users rationally specify the highest-stakes attributes first, while assistants prioritize asking on the highest-residual-stake attributes. The fundamental contract-theoretic mechanism, namely the substitution between upstream specification effort and automated clarification under misaligned asking friction, generalizes directly to the heterogeneous-loss setting.

A related limitation concerns the empirical identification of λ_A. Every bias-driven prediction (Corollary 2, Proposition 7, and predictions P2, P6, P7, and P8) is stated in terms of under-asking bias ($\lambda_A > c_Q$), but backing out λ_A from observed policy alone is circular. Lemma 1 formalizes this as an exact non-identification result: a^{SE} depends on (μ_A, λ_A) only through their ratio $\tilde{\rho} = (\lambda_A - c_Q)/\mu_A$, so a discounting-driven leader and a friction-driven leader with the identical excess friction ratio cannot be distinguished from pooling behavior alone. An independent empirical handle, such as an explicit penalty on turn count in the training objective or reward specification, would be needed before the bias-versus-heterogeneity predictions in Section 8 can be tested without

circularity. The two-type restriction in Model II avoids ironing complications at the cost of generality; a continuum formulation using the standard Mirrlees technique (Mirrlees, 1971; Myerson, 1979) is a natural next step to characterize the continuous distribution of downward distortions across arbitrary cost distributions. We also assume truthful revelation of specified attributes (abstracting from cheap-talk ambiguity within what is communicated) and a single leader. Modeling competition among multiple assistants represents an appealing extension, as market competition could plausibly discipline asking bias directly.

A broader methodological conclusion concerns the limitations of applying classical contract-theoretic intuition directly to human–AI settings without accounting for non-monetary, partially aligned incentives. In classical mechanism design (such as Baron–Myerson, Laffont–Tirole, or costly verification models (Baron and Myerson, 1982; Laffont and Tirole, 1986; Ben-Porath et al., 2014)), the principal pays monetary transfers to extract surplus, and agent informational rent directly reduces the principal's payoff. In human–AI interaction, the assistant's objective directly incorporates user welfare. As a consequence, incentive compatibility is costless in the absence of bias (Proposition 6), participation constraints do not bind at the bottom of the type distribution, and unbiased pooling naturally induces corner solutions rather than interior compromises (Corollary 4). Economic analyses of automated assistance must therefore explicitly incorporate the non-monetary, partially aligned nature of the interaction rather than relying on direct analogies to cash-transfer mechanisms.

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